

The Annotated Rudin

Principles of Mathematical Analysis, Third Edition

Walter Rudin

CHAPTER 1

The Real and Complex Number Systems

Rudin's text appears here complete and unaltered — his sentences, his paragraph breaks, his numbering. Nothing has been paraphrased away. The annotation is strictly additive: numbered markers in his prose key to notes in the outer margin, and longer commentary sits in ruled boxes between his paragraphs, where it can be skipped without breaking his argument.

How to read this book

Everything set in the main column below is Rudin, verbatim. You can read it straight through and never look at an annotation. The annotation layer sits in two places, both outside his text.

Margin notes. A small superscript number in Rudin's text — like this¹ — keys to a numbered note in the outer margin. Since the book prints in black and white, the *tier* of a note is carried by how its number is set:

¹ *The marker is the only mark added to his page, and it is the same for every tier so his line spacing is undisturbed. The number out here tells you what kind of note it is.*

7

Plain number — unpacking. What the sentence is actually saying, where a formula came from, an omitted algebraic step.

7

Outlined number — going deeper. Context beyond the course: what breaks in general, which later theorem this sets up. Skippable.

7

Reversed number — warning. A specific belief students form here that is false, or a place where Rudin's wording will mislead you.

Interleaved boxes. When a note runs longer than a margin will hold, it becomes a box between Rudin's paragraphs, framed by the same three tiers: a **solid** left rule for unpacking, a **dashed** left rule for going deeper, and rules **top and bottom** with a black tab for a warning.

Every figure is an addition. Rudin's Chapter 1 contains no diagrams at all. Any picture you see here is annotation, drawn to expose geometry that his algebra states without illustrating; each carries a grey italic caption.

Where the text is corrected. The main column follows a retyped edition whose own foreword concedes that fresh typos were probably introduced. Where one is evident, the reading is restored from Rudin's third edition and a warning-tier note records the change — silent emendation would make “verbatim” meaningless.

Structural commentary. Two further boxes carry no mathematics at all. They explain the book rather than its contents — why a chapter exists, why Rudin ordered it as he did, why a section that looks like a detour is not one.

HEAVY FRAME, BLACK TITLE BAR

Opens and closes every chapter: what it is for, how it is organized and why in that order, which parts are load-bearing. Where Rudin has explained his own choices — usually in the Preface — these boxes quote him rather than guess.

THIN FRAME, INLINE HEADING | Opens a section: why this material sits here, and what to hold on to.

Books cited in the notes. Where an annotation says another treatment does something differently, it means one of these. They are named because seeing the same theorem set up two ways is often what makes Rudin’s choice legible.

- | | |
|-------------------------------|--|
| <i>Tao, Analysis I–II</i> | Terence Tao, 3rd ed. Builds $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ from the Peano axioms and constructs $\mathbb{R}$ by Cauchy sequences rather than cuts. Volume II covers metric spaces and uniform convergence — Rudin’s Chapters 2, 4 and 7. |
| <i>Yu Pin, Tsinghua notes</i> | 数学分析讲义, Tsinghua / Yau Mathematical Sciences Center. Axiomatizes $\mathbb{R}$ by field, order, archimedean and <i>nested interval</i> axioms, then derives the supremum principle. |
| <i>ECNU text</i> | 数学分析 (Mathematical Analysis), 5th ed., 2018, East China Normal University. Standard Chinese two-volume text. Its §7.1 collects six equivalent formulations of completeness and proves them interderivable. |
| <i>USTC notes</i> | 中国科学技术大学少年班数学分析讲义 (程艺 et al., 2005). Teaches the main thread of calculus first and defers real-number theory to a later volume; lists Rudin among its three principal references. |

Chapter 1

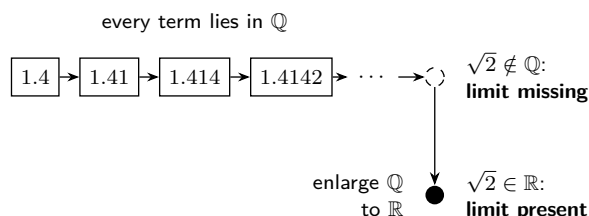
THE REAL AND COMPLEX NUMBER SYSTEMS

WHY THIS CHAPTER EXISTS

Consider the sequence

$$1, \quad 1.4, \quad 1.41, \quad 1.414, \quad 1.4142, \quad \dots$$

Every number in this sequence is rational. The sequence approaches $\sqrt{2}$, but $\sqrt{2}$ is not rational. Therefore the sequence has no limit that belongs to $\mathbb{Q}$.



The real numbers fill this gap by including $\sqrt{2}$. Rudin secures the property that rules out such gaps before using it throughout analysis — he states it as the *least-upper-bound property* (1.10), and it is what the word *complete* will name in 3.12.

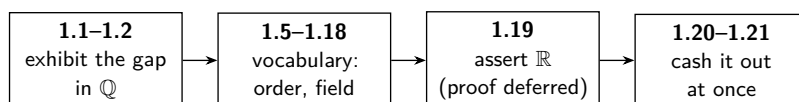
Everything asserting that something *exists* depends on it: that a bounded monotone sequence converges (3.14), that a continuous function on $[a, b]$ attains its bounds (4.16) and takes every intermediate value (4.23), that a continuous function is integrable (6.8). Each fails, or loses its meaning, over $\mathbb{Q}$ — $x^2 - 2$ changes sign on $[1, 2] \cap \mathbb{Q}$ without ever vanishing. Purely algebraic results survive; the existence theorems do not.

HOW IT IS ORGANIZED, AND WHY IN THAT ORDER

Rudin's own account. The obvious order would be to construct $\mathbb{R}$ from $\mathbb{Q}$ first and then do analysis on it. He declines, and says why in the Preface: beginning with the construction is “pedagogically unsound (though logically correct),” because “most students simply fail to appreciate the need for doing this.” Accordingly $\mathbb{R}$ “is introduced as an ordered field with the least-upper-bound

property,” and Dedekind’s construction is moved to the appendix, “where it may be studied and enjoyed whenever the time seems ripe.”

Reading that off the chapter, the shape is one designed to make you feel the need before paying the cost:



Four consequences are worth knowing in advance.

The construction is missing on purpose. Theorem 1.19 is stated and not proved; nothing in Chapters 2–11 depends on having read the appendix.

The abstraction comes before the object. Ordered sets (1.5–1.11) and fields (1.12–1.18) are defined *in general* before $\mathbb{R}$ is named, which lets Rudin characterise $\mathbb{R}$ by properties rather than by construction. He gives the second half of the reason himself at 1.13(c): proving the elementary consequences once, for an arbitrary field, means “we will not need to do it again for the real numbers and for the complex numbers.”

Two of the sections are equipment. The complex field (1.24–1.35) and Euclidean space (1.36–1.38) prove nothing about $\mathbb{R}$; they supply objects Chapters 2–11 need. The one result among them with real downstream weight is the Schwarz inequality, 1.35, which yields the metric on $\mathbb{R}^k$ and reappears for integrals in Exercise 6.10 (Hölder’s inequality, whose case $p = q = 2$ is there named Schwarz) and for square-integrable functions in 11.35.

Where the difficulty lies — a reader’s judgement, not Rudin’s. Not in 1.19, which is asserted, nor in the field axioms, which are routine. It is in 1.21 and in the appendix, the two places where a real number is produced out of nothing but a supremum. Follow those two arguments and you have the chapter.

HOW THE PROOFS IN THIS CHAPTER ARE FOUND

A proof you can only verify is a proof you cannot reproduce. Every proof in this chapter is built from one of seven discovery patterns, catalogued here once; the box at each proof works the details, and the same patterns carry most of Chapters 2 and 3.

1. Unwind the definition. When the objects are abstract, the axioms are the entire arsenal. Ask which axiom can *produce or*

remove the symbol standing between you and the goal, and the proof assembles itself (1.14, 1.15, 1.16, 1.18).

2. Hunt an invariant. To prove something impossible, find a quantity the two sides of the supposed equation cannot share — parity in 1.1, an integer strictly between 0 and 1 in 3.32, strict decrease in Tennenbaum’s dissection.

3. Transport the problem to a supremum. Completeness is the chapter’s only existence tool, so a proof that must *produce* a number has one move: build a set whose supremum is the number wanted — L in 1.11, $\{nx\}$ in 1.20(a), E in 1.21 — then interrogate the supremum with the two clauses of 1.8: step back to find members, step past to find bounds.

4. Work backwards for the constants. Every mysterious choice (h in 1.21, the coefficients B, C in 1.35) is found by writing down the target inequality and solving for the hypothesis, then presenting the steps in reverse.

5. Prove identities through uniqueness. To show two expressions are equal, show both answer a question that has exactly one answer (Corollary to 1.21, 1.33(c)).

6. Square away the absolute values. Statements about $|\cdot|$ become inner-product algebra after squaring; expand, bound the cross term, take roots at the end (1.33(e), 1.35, 1.37(e)).

7. Recognise pure verification. Some proofs (1.25–1.29, 1.31, 1.37 (a)–(c), most of the appendix) contain no idea to find — the idea lives in the *definition* being verified, and the proof is bookkeeping. Knowing that nothing is hidden is itself the skill: do not search a verification for insight that was spent earlier, at the definition.

INTRODUCTION

A satisfactory discussion of the main concepts of analysis (such as convergence, continuity, differentiation, and integration) must be based on an accurately defined number concept. We shall not, however, enter into any discussion of the axioms that govern the arithmetic of the integers, but assume familiarity with the rational numbers (i.e., the numbers of the form m/n where m and n are integers and $n \neq 0$).²

The rational number system is inadequate for many purposes, both as a field and as an ordered set. (These terms will be defined in Definitions 1.6 and 1.12.) For instance, there is no rational p such that $p^2 = 2$. (We shall prove this presently.) This leads to the introduction of so-called “irrational numbers” which are often written as infinite decimal expansions and are considered to be “approximated” by the

² Rudin is choosing a starting line, and he defends the choice in the Preface: beginning with a construction of $\mathbb{R}$ is “pedagogically unsound (though logically correct)” because “most students simply fail to appreciate the need for doing this.” So the construction is exiled to the appendix, and $\mathbb{R}$ will arrive in 1.19 as an assertion. This sentence marks everything he is willing to take for granted.

corresponding finite decimals. Thus the sequence

$$1, 1.4, 1.41, 1.414, 1.4142, \dots$$

“tends to $\sqrt{2}$.”³ But unless the irrational number $\sqrt{2}$ has been clearly defined, the question must arise: Just what is it that this sequence “tends to”?

This sort of question can be answered as soon as the so-called “real number system” is constructed.

1.1 Example. We now show that the equation

$$p^2 = 2 \tag{1.1}$$

is not satisfied by any rational p . If there were such a p , we could write $p = m/n$ where m and n are integers that are not both even.⁴ Let us assume this is done. Then (1.1) implies

$$m^2 = 2n^2, \tag{1.2}$$

This shows that m^2 is even. Hence m is even (if m were odd, m^2 would be odd), and so m^2 is divisible by 4. It follows that the right side of (1.2) is divisible by 4, so that n^2 is even, which implies that n is even.

The assumption that (1.1) holds thus leads to the conclusion that both m and n are even, contrary to our choice of m and n . Hence (1.1) is impossible for rational p .

HOW THE PARITY ARGUMENT IS FOUND

Faced with $m^2 = 2n^2$, the search is for an *invariant*: some property that the equation forces on both sides but that the two sides cannot share. The right side is visibly even, so ask what evenness of m^2 says about m — and the cheapest arithmetic invariant, parity, turns out to propagate: m even makes m^2 divisible by 4, pushing evenness onto n^2 and then n .

Two design choices become visible once you look for them. The normalisation “not both even” exists so that the invariant has something to contradict — without it, “both even” is not absurd, merely reducible. And the argument is really about the exponent of 2: each side of $m^2 = 2n^2$ contains the prime 2 an even number of times on the left and an odd number on the right, which is the version that generalises (Exercise 1.2 runs it with the prime 3 on $m^2 = 12n^2$). Parity is that count, read modulo 2.

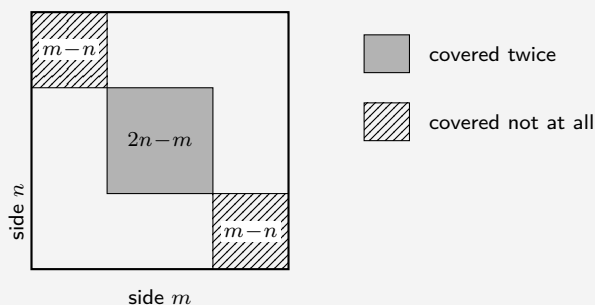
An alternative proof, done entirely by cutting up a square

Rudin argues by parity. There is a second proof, due to Stanley Tennenbaum, which uses no arithmetic beyond areas — and it makes the descent visible rather than hidden in the phrase “not both even.”

3 The scare quotes are doing real work. “Tends to” is undefined until Definition 3.1, and the thing it tends to is undefined until Theorem 1.19. Rudin is pointing at a circularity he is about to break, not relying on one.

4 Why we may assume this: put p in lowest terms. If m and n were both even we could cancel a factor of 2 and repeat, and the process must stop because $|m|$ strictly decreases. That appeal to “must stop” is well-ordering, so the step is not quite free. It can be avoided: Tao (Analysis I, 4.4.4) drops the assumption and instead turns each solution $p^2 = 2q^2$ into a smaller one, contradicting the principle of infinite descent. Same content, with the well-ordering moved into the open.

Suppose $m^2 = 2n^2$ with m, n positive integers. Then $n < m < 2n$, since $m^2 = 2n^2$ lies strictly between n^2 and $4n^2$. Draw a square of side m and lay a square of side n into two opposite corners. Because $m < 2n$ they overlap; because $m > n$ two corners are left bare.



Two squares of side n dropped into opposite corners of a square of side m .

The two n -squares have total area $2n^2$, which by hypothesis is exactly m^2 , the area of the big square. So whatever is counted twice must have the same area as whatever is missed:

$$(2n - m)^2 = 2(m - n)^2.$$

That is the same equation again, in strictly smaller positive integers — $0 < 2n - m < m$ and $0 < m - n < n$. Repeating produces an infinite strictly decreasing sequence of positive integers, which is impossible. Hence no such m, n exist.

The two proofs use the same principle. Rudin's “not both even” silently invokes well-ordering to stop a descent; here the descent is performed in the open, and each step is a picture.

We now examine this situation a little more closely. Let A be the set of all positive rationals p such that $p^2 < 2$ and let B consist of all positive rationals p such that $p^2 > 2$. We shall show that A contains no largest number and B contains no smallest.⁵

More explicitly, for every p in A we can find a rational q in A such that $p < q$, and for every p in B we can find a rational q in B such that $q < p$.

To do this, we associate with each rational $p > 0$ the number

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \quad (1.3)$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \quad (1.4)$$

If p is in A , then $p^2 - 2 < 0$, (1.3) shows that $q > p$ and (1.4) shows that $q^2 < 2$. Thus q is in A .

⁵ This, and not the irrationality of $\sqrt{2}$, is the real content of §1.1. Irrationality says $\sqrt{2} \notin \mathbb{Q}$. This says $\mathbb{Q}$ has a gap where $\sqrt{2}$ should be — a structural defect, and the one Definition 1.10 will name.

If p is in B , then $p^2 - 2 > 0$, (1.3) shows that $0 < q < p$, and (1.4) shows that $q^2 > 2$. Thus q is in B .

WHERE (1.3) COMES FROM

Rudin produces this formula with no derivation, and it is the first place in the book where a reader stalls. It is not Newton's method. Ask instead: which map $p \mapsto q$ has $\sqrt{2}$ as a *fixed point*? Try the simplest shape, a ratio of linear terms $q = (ap + b)/(cp + d)$. Setting $q = p$ gives $p(cp + d) = ap + b$, and we want that to collapse to $p^2 = 2$. The smallest-integer solution is $q = (2p + 2)/(p + 2)$: it gives $p^2 + 2p = 2p + 2$, that is, $p^2 = 2$.

Why it moves p toward $\sqrt{2}$ without crossing it becomes visible once you factor the error:

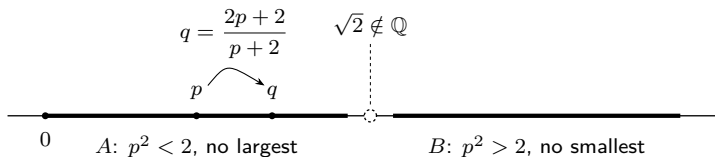
$$q - \sqrt{2} = \frac{2 - \sqrt{2}}{p + 2} (p - \sqrt{2}).$$

For $p > 0$ the multiplier lies strictly between 0 and $(2 - \sqrt{2})/2 \approx 0.29$. That it is *positive* keeps q on the same side of $\sqrt{2}$ as p — Rudin's " q is in A " and " q is in B ." That it is *less than one* makes the error shrink, so q is strictly better than p and no member of A can be largest.

Equation (1.4) is worth checking rather than accepting:

$$(2p + 2)^2 - 2(p + 2)^2 = (4p^2 + 8p + 4) - (2p^2 + 8p + 8) = 2p^2 - 4.$$

There is a second way to say what this establishes, worth knowing because it is the form other books use. Tao (*Analysis I*, 4.4.5) proves: for every rational $\varepsilon > 0$ there is a rational $x \geq 0$ with $x^2 < 2 < (x + \varepsilon)^2$. Rudin's version is about the sets A and B having no extreme member; Tao's is about how narrowly they can be straddled. They say the same thing — the gap is real, and it is narrower than any rational you care to name.



Every $p \in A$ is beaten by some $q \in A$, and every $p \in B$ beats some $q \in B$. The two sets crowd a point that is not there.

The estimate behind the contraction principle

That factorisation says $p \mapsto (2p + 2)/(p + 2)$ shrinks the distance to $\sqrt{2}$ by a factor of at most 0.29 at every step. It is the same estimate that Theorem 9.23, the contraction principle, later abstracts — though 9.23 itself does not apply here, since it needs a *complete* space and the positive rationals are exactly what is not complete.

That is the point: the limit $\sqrt{2}$ does not exist in $\mathbb{Q}$, which is why Rudin must argue by hand that no member of A is largest rather than simply passing to a fixed point.

Iterating from $p_0 = 1$ gives

$$1, \frac{4}{3}, \frac{7}{5}, \frac{24}{17}, \frac{41}{29}, \frac{140}{99}, \dots$$

Every term stays below $\sqrt{2}$, as the positive multiplier requires. Some are convergents of $\sqrt{2} = [1; 2, 2, 2, \dots]$ — $\frac{7}{5}$ and $\frac{41}{29}$ are — but not all: the convergents run $1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \dots$, and this iteration visits every second one.

Newton's method on $f(p) = p^2 - 2$ would give $p \mapsto (p^2 + 2)/(2p)$ instead: faster, but from any starting point it lands *above* $\sqrt{2}$, so it would prove “ B has no smallest” and say nothing about A . Rudin's map does both halves at once.

6 The property Rudin describes here — between any two elements lies a third — is not the property Definition 1.10 will demand, and this sentence is the first warning that the two are independent. $\mathbb{Q}$ has the first and lacks the second. $\mathbb{Z}$ has the second (every nonempty bounded-above set of integers has a largest element) and lacks the first. So no amount of “no visible gaps between neighbours” delivers completeness.

7 Rudin's $\subset$ permits equality — it is what most modern texts write $\subseteq$. He writes “proper subset” when he means strict containment. Read every $\subset$ in this book as $\subseteq$, or you will misread hypotheses for the next 300 pages.

1.2 Remark. The purpose of the above discussion has been to show that the rational number system has certain gaps, in spite of the fact that between any two rationals there is another: if $r < s$, then $r < (r + s)/2 < s$.⁶ The real number system fills these gaps. This is the principal reason for the fundamental role which it plays in analysis.

In order to elucidate its structure, as well as that of the complex numbers, we start with a brief discussion of the general concepts of *ordered set* and *field*.

Here is some of the standard set-theoretic terminology that will be used throughout this book.

1.3 Definitions. If A is any set (whose elements may be numbers or any other objects), we write $x \in A$ to indicate that x is a member (or an element) of A . If x is not a member of A , we write $x \notin A$.

The set which contains no element will be called the *empty set*. If a set has at least one element, it is called *nonempty*.

If A and B are sets, and if every element of A is an element of B , we say that A is a *subset* of B , and we write $A \subset B$, or $B \supset A$.⁷ If, in addition, there is an element of B which is not in A , then A is said to be a *proper subset* of B . Note that $A \subset A$ for every set A .

If $A \subset B$ and $B \subset A$, we write $A = B$. Otherwise, $A \neq B$.

1.4 Definition. Throughout Chap. 1, the set of all rational numbers will be denoted by $\mathbb{Q}$.

ORDERED SETS

WHY THIS SECTION IS HERE | Nothing in this section mentions $\mathbb{R}$, and on a first reading it feels like an unmotivated detour into ab-

straction. It is not. Rudin has just shown that $\mathbb{Q}$ fails in a specific way, and he now needs language precise enough to *name* that failure — because the name will become the defining property of $\mathbb{R}$ in 1.19. Definitions 1.5 to 1.7 are only scaffolding; 1.8 and 1.10 are the payload. If you read one thing here, read Definition 1.10, and notice that it is exactly what Example 1.9(a) showed $\mathbb{Q}$ does not have.

1.5 Definition. Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties:

- (i) If $x \in S$ and $y \in S$, then one and only one of the statements

$$x < y, \quad x = y, \quad y < x$$

is true.⁸

- (ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

The statement “ $x < y$ ” may be read as “ x is less than y ” or “ x is smaller than y ” or “ x precedes y ”.

It is often convenient to write $y > x$ in place of $x < y$.

The notation $x \leq y$ indicates that $x < y$ or $x = y$, without specifying which of these two is to hold. In other words, $x \leq y$ is the negation of $x > y$.⁹

1.6 Definition. An *ordered set* is a set S in which an order is defined.

For example, $\mathbb{Q}$ is an ordered set if $r < s$ is defined to mean that $s - r$ is a positive rational number.

1.7 Definition. Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E .

Lower bounds are defined the same way (with $\geq$ in place of $\leq$).

1.8 Definition. Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

- (i) α is an upper bound of E .
(ii) If $\gamma < \alpha$ then γ is not an upper bound of E .¹⁰

Then α is called the *least upper bound* of E [that there is at most one such α is clear from (ii)] or the *supremum* of E , and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E .

8 Trichotomy. “One and only one” packs in two separate demands: at least one holds, so the order is total; and at most one holds, so $<$ is irreflexive and antisymmetric. Drop totality and you get a partial order, where suprema can fail to exist for reasons having nothing to do with gaps.

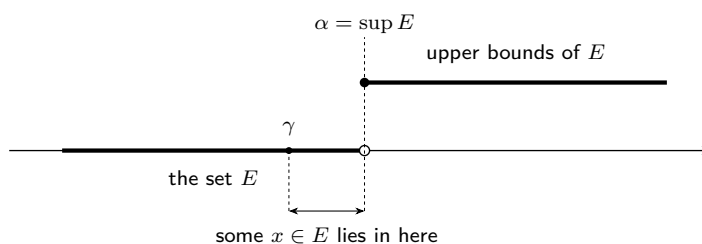
9 That last sentence is a small theorem, not a definition: “ $x < y$ or $x = y$ ” is equivalent to “not $x > y$ ” precisely because of trichotomy in (i). In a partial order the two come apart, which is why $\leq$ is normally taken as the primitive notion there.

10 This is the clause you will actually use, and it is stated contrapositively. The working form: for every $\gamma < \alpha$ there is an $x \in E$ with $x > \gamma$. In $\mathbb{R}$, equivalently: for every $\varepsilon > 0$ there is $x \in E$ with $x > \alpha - \varepsilon$. Reach for that sentence every time a proof opens “let $\alpha = \sup E$.”

WHY THE LEAST UPPER BOUND AND NOT THE LARGEST ELEMENT

Because the largest element usually does not exist, and the supremum is the repair. Take $E_1 = \{r \in \mathbb{Q} : r < 0\}$. It has no largest element: for any $r < 0$ there is $r/2$ with $r < r/2 < 0$. But the set of its upper bounds — everything ≥ 0 — does have a smallest one, namely 0.

The manoeuvre is to move the question from E , where the answer is often “none,” to the set of upper bounds of E , where it is often “yes.” Rudin’s bracketed aside is the payoff: uniqueness follows immediately from (ii), so speaking of “the” supremum is legitimate.



α is the seam. Everything to its right bounds E ; nothing to its left does.

The picture also shows why clause (ii) is the useful one. Stepping back from α by any amount lands you at a γ that is *not* an upper bound, which is to say some element of E sticks out past it. That is the sentence you will use.

1.9 Examples.

- (a) Consider the sets A and B of Example 1.1 as subsets of the ordered set $\mathbb{Q}$. The set A is bounded above. In fact, the upper bounds of A are exactly the members of B .¹¹ Since B contains no smallest member, A has no least upper bound in $\mathbb{Q}$.

Similarly, B is bounded below: the set of all lower bounds of B consists of A and of all $r \in \mathbb{Q}$ with $r \leq 0$. Since A has no largest member, B has no greatest lower bound in $\mathbb{Q}$.

- (b) If $\alpha = \sup E$ exists, then α may or may not be a member of E . For instance, let E_1 be the set of all $r \in \mathbb{Q}$ with $r < 0$. Let E_2 be the set of all $r \in \mathbb{Q}$ with $r \leq 0$. Then

$$\sup E_1 = \sup E_2 = 0,$$

$$\text{and } 0 \notin E_1, 0 \in E_2.^{12}$$

- (c) Let E consist of all numbers $1/n$, where $n = 1, 2, 3, \dots$. Then $\sup E = 1$, which is in E , and $\inf E = 0$, which is not in E .

1.10 Definition. An ordered set S is said to have the *least-upper-bound property* if the following is true:

¹¹ Not exactly, and Rudin is being loose. Every $\beta \in B$ bounds A , and no member of A does. But one must also rule out a positive rational with $\beta^2 = 2$, and that is impossible only by Example 1.1. The claim holds, but it leans on the irrationality result proved a page earlier.

¹² Both suprema exist in $\mathbb{Q}$. Part (a) fails not because $\mathbb{Q}$ is too small in general but because of where that particular gap sits. Whether $\sup E$ lies in E is a separate question from whether $\sup E$ exists, and running the two together is the most common first-week error.

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S .¹³

Example 1.9(a) shows that $\mathbb{Q}$ does not have the least-upper-bound property.

Rudin's choice of axiom is one of several

Completeness can be installed in an ordered field in several equivalent ways, and which one a book takes as primitive shapes everything after it.

Rudin takes the least-upper-bound property as the defining property (1.10), asserts a field with it (1.19), and *derives* the archimedean property from it (1.20a).

Yu Pin (Tsinghua) axiomatises $\mathbb{R}$ by field, order, archimedean, and the *nested interval* axiom — every decreasing sequence of closed bounded intervals has a common point. He observes that $\mathbb{Q}$ satisfies the first three and fails only the fourth, then derives the supremum principle from it. Archimedes must be assumed separately, because nested intervals alone does not imply it; he proves the converse too, so over an archimedean ordered field the two are equivalent.

Tao (*Analysis I*) assumes nothing, builds $\mathbb{R}$ from $\mathbb{Q}$ as equivalence classes of Cauchy sequences, and *proves* the least-upper-bound property as a theorem.

The ECNU text lays out the whole menu. Its §7.1 collects six statements — the supremum principle, monotone convergence, nested intervals, Heine–Borel, Bolzano–Weierstrass, and the Cauchy criterion — and proves the cycle $1 \Rightarrow 2 \Rightarrow \cdots \Rightarrow 6 \Rightarrow 1$, so that any one of them may serve as the axiom and the other five follow.

That last point is worth carrying into Chapter 2. Rudin does not present these as six faces of one property; he proves them separately and far apart — 2.41 and 2.42 for compactness and limit points in $\mathbb{R}^k$, 3.11 for Cauchy sequences, 3.14 for monotone ones. Each is a theorem earned from 1.19. Knowing they are interderivable tells you that those later proofs are not accumulating new strength about $\mathbb{R}$: they are unpacking, in different vocabularies, the single property assumed here.

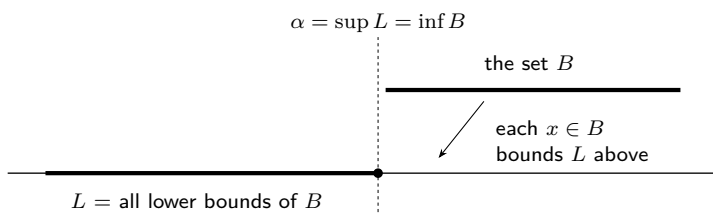
13 Both hypotheses are load-bearing, and dropping either breaks the definition rather than merely weakening it. In $\mathbb{R}$: every real is an upper bound of $\emptyset$, so $\emptyset$ has no least one; and $\mathbb{Z}$ is unbounded above, so it has none either. Any proof that produces a supremum owes you a check of both.

We shall now show that there is a close relation between greatest lower bounds and least upper bounds, and that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

THE PICTURE BEHIND THEOREM 1.11

We are handed a hypothesis about suprema and asked to produce an infimum, so the only move available is to find a set whose *supremum* is the infimum we want. The set of lower bounds L is that set —

it sits entirely below B , every element of B bounds it from above, and the two meet at a single point.



The two sets share one boundary point, and the theorem says that point is both $\sup L$ and $\inf B$.

1.11 Theorem. Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B . Then

$$\alpha = \sup L$$

exists in S and $\alpha = \inf B$.

In particular, $\inf B$ exists in S .

Proof. Since B is bounded below, L is not empty. Since L consists of exactly those $y \in S$ which satisfy the inequality $y \leq x$ for every $x \in B$, we see that every $x \in B$ is an upper bound of L .¹⁴ Thus L is bounded above. Our hypothesis about S implies therefore that L has a supremum in S ; call it α .

If $\gamma < \alpha$ then (see Definition 1.8) γ is not an upper bound of L , hence $\gamma \notin B$.¹⁵ It follows that $\alpha \leq x$ for every $x \in B$. Thus $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L .

We have shown that $\alpha \in L$ but $\beta \notin L$ if $\beta > \alpha$. In other words, α is a lower bound of B , but β is not if $\beta > \alpha$. This means that $\alpha = \inf B$. $\square$

HOW THE PROOF FLOWS

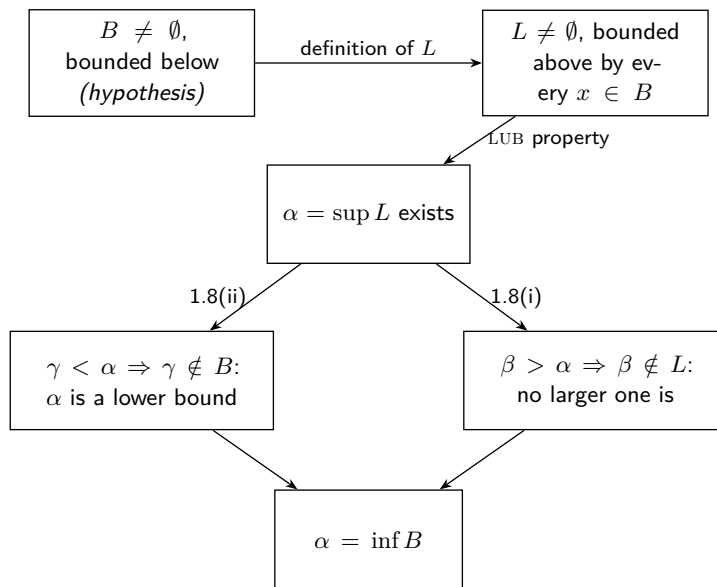
- 1. Choose the method.** The hypothesis speaks of suprema, the goal of an infimum, and the least-upper-bound property is the only existence tool in the chapter. So the method is forced — pattern 3 of the strategy index: *find a set whose supremum is the number you want*, then take its sup. The candidate is L , the set of all lower bounds of B ; the picture before the theorem shows why it is right: L ends exactly where B begins.
- 2. Buy the supremum.** The lub property sells only to sets that are nonempty and bounded above, so both tickets must be produced. $L \neq \emptyset$ is the hypothesis “ B is bounded below” restated. L bounded above is the proof’s one genuine observation: every $x \in B$ bounds L , by the very definition of “lower bound.”

14 This single sentence is the whole idea. We are handed a hypothesis about suprema and asked for an infimum, so we must find a set whose supremum is the infimum we want. L is that set: B presses down on it from above, and the two meet at one point.

15 Compressed to the point of opacity. Unpacked: γ failing to be an upper bound of L produces some $y \in L$ with $y > \gamma$. Since y is a lower bound of B , every $x \in B$ satisfies $x \geq y > \gamma$. So γ itself cannot belong to B .

Payment accepted: $\alpha = \sup L$ exists.

3. **Show α is a lower bound** (first clause of the goal). Rudin argues by testing every $\gamma < \alpha$: clause (ii) of 1.8 says γ is not an upper bound of L , so some $y \in L$ exceeds γ ; since y underlies all of B , so does γ , strictly — hence $\gamma \notin B$. Nothing in B lies below α , which is precisely $\alpha \in L$.
4. **Show nothing larger works** (second clause). Any $\beta > \alpha$ exceeds the supremum of L , hence $\beta \notin L$: it fails to be a lower bound. This step uses clause (i) of 1.8 — α is an upper bound of L .
5. **Read off the conclusion.** Steps 3 and 4 are exactly the two clauses of Definition 1.8, mirrored for the infimum, so $\alpha = \inf B$. Note the audit: clause (ii) of the supremum was spent in step 3, clause (i) in step 4 — each exactly once, the usual sign that L was the right set to build (compare the same remark at 1.21).



The proof as a flow diagram: hypotheses qualify L , the least-upper-bound property mints α , and the two clauses of Definition 1.8 each certify one half of “ $\alpha = \inf B$.”

Why this is not a triviality

The argument runs equally well in reverse, so for ordered sets the least-upper-bound property and the greatest-lower-bound property are *equivalent*. That is why Rudin never states a separate greatest-lower-bound axiom for $\mathbb{R}$: Theorem 1.19 gives him one and Theorem 1.11 hands him the other.

Every later appearance of $\inf$ rests on this — lower Riemann–Stieltjes sums in 6.2, $\liminf$ in 3.16, the distance from a point to a set in the exercises. It is a two-paragraph theorem that the rest of

the book quietly leans on.

FIELDS

WHY THIS SECTION IS HERE | This is the second half of the vocabulary, and the less interesting half. An ordered set gave Rudin the language for *gaps*; a field gives him the language for *arithmetic*. $\mathbb{R}$ will be defined in 1.19 as an object having both, plus one extra property, so both notions must be on the table first.

Rudin tells you outright how to read 1.14–1.18: they are “some of the familiar properties of $\mathbb{Q}$ ” rederived from the axioms so that “once we do this, we will not need to do it again for the real numbers and for the complex numbers.” That is the entire point. Nothing here will surprise you; the value is that every one of these facts is now available for $\mathbb{R}$, for $\mathbb{C}$, and for any other field, purchased once. Skim the proofs, note Definition 1.17, and move on — but see the warning after 1.18, because the ordering axioms are where students later import assumptions that are not there.

1.12 Definition. A *field* is a set F with two operations, called *addition* and *multiplication*, which satisfy the following so-called “field axioms” (A), (M), and (D):

(A) Axioms for addition

- (A1) If $x \in F$ and $y \in F$, then their sum $x + y$ is in F .
- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in F$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- (A4) F contains an element 0 such that $0 + x = x$ for every $x \in F$.
- (A5) To every $x \in F$ corresponds an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

- (M1) If $x \in F$ and $y \in F$, then their product xy is in F .¹⁶
- (M2) Multiplication is commutative: $xy = yx$ for all $x, y \in F$.
- (M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in F$.
- (M4) F contains an element $1 \neq 0$ such that $1x = x$ for every $x \in F$.¹⁷
- (M5) If $x \in F$ and $x \neq 0$ then there exists an element $1/x \in F$ such that

$$x \cdot (1/x) = 1.$$

16 Source correction. The retyped edition this book follows drops “and $y \in F$ ” here, and drops the opening “If” from (A1); its own foreword warns that new typos were likely introduced. Both clauses are restored above from Rudin’s third edition. Without $y \in F$ the axiom does not say what it must — that F is closed under multiplication.

17 The clause $1 \neq 0$ is not pedantry. Without it the one-element set $\{0\}$ satisfies every other axiom, and then Proposition 1.16(a) would make every element equal to 0. Excluding it is what keeps “field” from being vacuous.

(D) The distributive law

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in F$.¹⁸

1.13 Remarks.

- (a) One usually writes (in any field)

$$\begin{array}{ccccccc} x - y, & \frac{x}{y}, & x + y + z, & xyz, & & & \\ x^2, & x^3, & 2x, & 3x, \dots & & & \end{array}$$

in place of

$$\begin{array}{ccccccc} x + (-y), & x \cdot \left(\frac{1}{y}\right), & (x + y) + z, & (xy)z, & & & \\ xx, & xxx, & x + x, & x + x + x, \dots & & & \end{array}$$

- (b) The field axioms clearly hold in $\mathbb{Q}$, the set of all rational numbers, if addition and multiplication have their customary meaning. Thus $\mathbb{Q}$ is a field.
- (c) Although it is not our purpose to study fields (or any other algebraic structures) in detail, it is worthwhile to prove that some of the familiar properties of $\mathbb{Q}$ are consequences of the field axioms; once we do this, we will not need to do it again for the real numbers and for the complex numbers.

1.14 Proposition. *The axioms for addition imply the following statements.*

- (a) *If $x + y = x + z$ then $y = z$.*
- (b) *If $x + y = x$ then $y = 0$.*
- (c) *If $x + y = 0$ then $y = -x$.*
- (d) $-(-x) = x$.

Statement (a) is a cancellation law. Note that (b) asserts the uniqueness of the element whose existence is assumed in (A4), and that (c) does the same for (A5).

Proof. If $x + y = x + z$, the axioms (A) give

$$\begin{aligned} y &= 0 + y = (-x + x) + y = -x + (x + y) \\ &= -x + (x + z) = (-x + x) + z = 0 + z = z. \end{aligned}$$

This proves (a). Take $z = 0$ in (a) to obtain (b). Take $z = -x$ in (a) to obtain (c). Since $-x + x = 0$, (c) (with $-x$ in place of x) gives (d). $\square$

[18] (D) is the only axiom linking the two operations, and every proof below that mixes $+$ and $\cdot$ must route through it. Watch 1.16(a): the fact that $0x = 0$ — which looks like arithmetic so basic it needs no proof — is derivable only because (D) exists.

THE PATTERN OF ALL FOUR PROOFS

Every step in that display is a single named axiom, and it is worth naming them once: $0 + y = y$ is (A4); $0 = -x + x$ is (A5) with (A2); the regrouping is (A3). Nothing else is used — in particular, no multiplication and no order.

The economy of (b), (c), (d) is the point. Once cancellation (a) is available, the rest are not new arguments but *substitutions* into it. Rudin's "take $z = 0$ " and "take $z = -x$ " are complete proofs, and the habit of proving one general statement and specialising it repeatedly is one he will use constantly.

How is the chain for (a) *found*, rather than checked? Work from the goal. You must delete x from $x + y$; the only axiom that annihilates anything is (A5), so $-x$ must enter; the only license to bring it next to x is regrouping, (A3); and once $-x + x$ becomes 0, (A4) removes it. Each step is forced — the axioms offer exactly one tool per obstacle, so the proof writes itself left to right once you know what must disappear. This is pattern 1 of the strategy index: with abstract objects, unwind the definitions, because there is nothing else to use.

1.15 Proposition. *The axioms for multiplication imply the following statements.*

- (a) *If $x \neq 0$ and $xy = xz$ then $y = z$.*
- (b) *If $x \neq 0$ and $xy = x$ then $y = 1$.*
- (c) *If $x \neq 0$ and $xy = 1$ then $y = 1/x$.*
- (d) *If $x \neq 0$ then $1/(1/x) = x$.*

The proof is so similar to that of Proposition 1.14 that we omit it.¹⁹

1.16 Proposition. *The field axioms imply the following statements, for any $x, y, z \in F$.*

- (a) $0x = 0$.
- (b) *If $x \neq 0$ and $y \neq 0$ then $xy \neq 0$.*
- (c) $(-x)y = -(xy) = x(-y)$.
- (d) $(-x)(-y) = xy$.

Proof. $0x + 0x = (0 + 0)x = 0x$. Hence Proposition 1.14(b) implies that $0x = 0$, and (a) holds.²⁰ Next, assume $x \neq 0$, $y \neq 0$, but $xy = 0$. Then

$$1 = \left(\frac{1}{y}\right) \left(\frac{1}{x}\right) xy = \left(\frac{1}{y}\right) \left(\frac{1}{x}\right) 0 = 0,$$

a contradiction. Thus (b) holds.

The first equality in (c) comes from

$$(-x)y + xy = (-x + x)y = 0y = 0,$$

¹⁹ This is Exercise 1.3. Copy the proof of 1.14 with $+$ $\rightarrow \cdot$, $0 \rightarrow 1$, $-x \rightarrow 1/x$, and carry the hypothesis $x \neq 0$ along; the axioms (M1)–(M5) match (A1)–(A5) exactly except for that restriction. The restriction is unavoidable: (M5) grants inverses only to nonzero elements, which is precisely why the multiplicative statements need a hypothesis the additive ones did not.

²⁰ The trick in one line: manufacture an equation of the form $u + u = u$, then quote 1.14(b) to conclude $u = 0$. How is it found? Ask what the axioms know about 0: only (A4), $0 = 0 + 0$. So substitute that and distribute — the sole fact about 0 meets the sole axiom linking $+$ to $\cdot$, and the proof is the only sentence one can write. First place (D) does any work.

combined with Proposition 1.14(c); the other half of (c) is proved in the same way. Finally,

$$(-x)(-y) = -[x(-y)] = -[-(xy)] = xy$$

by (c) and Proposition 1.14(d). $\square$

What (b) actually says

Part (b) is the statement that a field has *no zero divisors*, and unlike the rest of this proposition it is not a formality. There are perfectly good rings where it fails: in the integers mod 6, $2 \cdot 3 = 0$ with neither factor zero. What rules that out here is (M5) — the existence of inverses — and the proof shows exactly how: multiply by $1/x$ and $1/y$ and the supposed $xy = 0$ collapses 1 to 0, contradicting $1 \neq 0$ in (M4).

This is why $\mathbb{Z}/6\mathbb{Z}$ is not a field while $\mathbb{Z}/5\mathbb{Z}$ is, and it is quietly required every time a later argument divides by a nonzero quantity.

1.17 Definition. An *ordered field* is a field F which is also an ordered set, such that

- (i) $x + y < x + z$ if $x, y, z \in F$ and $y < z$,
- (ii) $xy > 0$ if $x \in F$, $y \in F$, $x > 0$, and $y > 0$.

If $x > 0$, we call x *positive*; if $x < 0$, x is *negative*.²¹

For example, $\mathbb{Q}$ is an ordered field.

All the familiar rules for working with inequalities apply in every ordered field: Multiplication by positive [negative] quantities preserves [reverses] inequalities, no square is negative, etc. The following proposition lists some of these.

1.18 Proposition. *The following statements are true in every ordered field.*

- (a) *If $x > 0$ then $-x < 0$, and vice versa.*
- (b) *If $x > 0$ and $y < z$ then $xy < xz$.*
- (c) *If $x < 0$ and $y < z$ then $xy > xz$.*
- (d) *If $x \neq 0$ then $x^2 > 0$. In particular, $1 > 0$.*
- (e) *If $0 < x < y$ then $0 < 1/y < 1/x$.*

Proof.

- (a) If $x > 0$ then $0 = -x + x > -x + 0$, so that $-x < 0$. If $x < 0$ then $0 = -x + x < -x + 0$, so that $-x > 0$. This proves (a).
- (b) Since $z > y$, we have $z - y > y - y = 0$, hence $x(z - y) > 0$, and therefore

$$xz = x(z - y) + xy > 0 + xy = xy.$$

²¹ Only two axioms, and they are the minimum needed to tie the order to the arithmetic: translation preserves order, and positives are closed under multiplication. Everything else you believe about inequalities — 1.18(a)–(e), and the whole of Chapter 1's estimating — is derived from these two. Rudin is not listing convenient facts; he is exhibiting a complete basis.

(c) By (a), (b), and Proposition 1.16(c),

$$-[x(z - y)] = (-x)(z - y) > 0,$$

so that $x(z - y) < 0$, hence $xz < xy$.

(d) If $x > 0$, part (ii) of Definition 1.17 gives $x^2 > 0$. If $x < 0$, then $-x > 0$, hence $(-x)^2 > 0$. But $x^2 = (-x)^2$ by Proposition 1.16(d). Since $1 = 1^2$, $1 > 0$.

(e) If $y > 0$ and $v \leq 0$, then $yv \leq 0$. But $y \cdot (1/y) = 1 > 0$. Hence $1/y > 0$. Likewise, $1/x > 0$. If we multiply both sides of the inequality $x < y$ by the positive quantity $(1/x)(1/y)$, we obtain $1/y < 1/x$.

□

THE ONE MOVE USED IN EVERY PART

Each proof converts an inequality into a statement about something being positive, applies an axiom, and converts back. Concretely: $y < z$ becomes $z - y > 0$ by (i) of 1.17, then (ii) applies, then (i) restores the inequality.

That is the whole technique. It is worth watching in (b), where it is cleanest: $z > y \Rightarrow z - y > 0 \Rightarrow x(z - y) > 0 \Rightarrow xz > xy$. Part (c) is (b) applied to $-x$, and (e) is (b) applied with the positive multiplier $(1/x)(1/y)$. Only (d) does something different, splitting into cases on the sign of x .

PART (D) IS THE SOURCE OF MOST OF CHAPTER 1'S LEVERAGE

$x^2 > 0$ for $x \neq 0$ looks too obvious to be worth stating. It is the reason $\mathbb{C}$ cannot be ordered (Exercise 1.8: $-1 = i^2$ would have to be both positive and negative), the reason $a^2 + b^2 > 0$ in the proof of (M5) for $\mathbb{C}$ at 1.25, and the reason $|z| = (z\bar{z})^{1/2}$ makes sense at 1.32.

The corollary $1 > 0$ deserves separate notice. It is not an axiom, and in a field with no order there is no sense in which 1 exceeds 0. Here it is *derived*, from $1 = 1^2$ — and once you have it, (i) of 1.17 gives $0 < 1 < 2 < 3 < \dots$, so every ordered field contains a copy of $\mathbb{N}$ in the right order. That fact is used without comment in the proof of 1.20.

THE REAL FIELD

THE HINGE OF THE CHAPTER | Everything so far has been preparation, and everything after is consequence. This section contains the one theorem the chapter was written to state.

Watch what Rudin does with it. He states 1.19, declines to prove it, and then immediately proves 1.20 and 1.21 — results he could have extracted from the construction “with very little extra effort,” as he says. He derives them from the least-upper-bound property

instead, on purpose: he wants you to see the property doing work before you see where it came from. Read 1.20 and 1.21 less as facts about $\mathbb{R}$ than as two worked demonstrations of how a supremum is used.

We now state the existence theorem which is the core of this chapter.

1.19 Theorem. *There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property.*

Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

The second statement means that $\mathbb{Q} \subset \mathbb{R}$ and that the operations of addition and multiplication in $\mathbb{R}$, when applied to members of $\mathbb{Q}$, coincide with the usual operations on rational numbers; also, the positive rational numbers are positive elements of $\mathbb{R}$.²² The members of $\mathbb{R}$ are called *real numbers*.

The proof of Theorem 1.19 is rather long and a bit tedious and is therefore presented in an Appendix to Chap. 1. The proof actually constructs $\mathbb{R}$ from $\mathbb{Q}$.²³

The next theorem could be extracted from this construction with very little extra effort. However, we prefer to derive it from Theorem 1.19 since this provides a good illustration of what one can do with the least-upper-bound property.

1.20 Theorem.

- (a) *If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that*

$$nx > y.$$

- (b) *If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.*

Part (a) is usually referred to as the *archimedean property* of $\mathbb{R}$. Part (b) may be stated by saying that $\mathbb{Q}$ is *dense* in $\mathbb{R}$: Between any two real numbers there is a rational one.²⁴

Proof. (a) Let A be the set of all nx , where n runs through the positive integers. If (a) were false, then y would be an upper bound of A . But then A has a least upper bound in $\mathbb{R}$. Put $\alpha = \sup A$. Since $x > 0$, $\alpha - x < \alpha$, and $\alpha - x$ is not an upper bound of A . Hence $\alpha - x < mx$ for some positive integer m . But then $\alpha < (m+1)x \in A$, which is impossible, since α is an upper bound of A .

(b) Since $x < y$, we have $y - x > 0$, and (a) furnishes a positive integer n such that

$$n(y - x) > 1.$$

Apply (a) again, to obtain positive integers m_1 and m_2 such that $m_1 > nx$, $m_2 > -nx$. Then

$$-m_2 < nx < m_1.$$

22 “Subfield” is a stronger claim than “subset.” It says the copy of $\mathbb{Q}$ sitting inside $\mathbb{R}$ has the arithmetic and the order it had before — so nothing you already know about rationals is invalidated by the enlargement.

23 Worth pausing on. Everything in the next ten chapters is a consequence of this one theorem plus the field and order axioms. If you accept 1.19, you never need the appendix; if you doubt it, the appendix settles it by exhibiting $\mathbb{R}$ as the set of Dedekind cuts of $\mathbb{Q}$. Either way the rest of the book is unaffected, which is why Rudin can safely postpone it.

24 Part (a) is a theorem, not an axiom, and it is not automatic. There are ordered fields in which it fails — fields containing an element larger than every integer. Such fields cannot have the least-upper-bound property, and (a) is precisely the proof of that.

Hence there is an integer m (with $-m_2 \leq m \leq m_1$) such that

$$m - 1 \leq nx < m.$$

If we combine these inequalities, we obtain

$$nx < m \leq 1 + nx < ny.$$

Since $n > 0$, it follows that

$$x < \frac{m}{n} < y.$$

This proves (b), with $p = m/n$. □

PART (A): GRAB THE ONLY HANDLE

How would one find this proof? Negate the conclusion: $nx \leq y$ for every n . That is a *boundedness* statement, and the only machinery 1.19 offers is that bounded sets have suprema — so the proof is forced to form $A = \{nx\}$ and take $\alpha = \sup A$. There is nothing else to try.

The attack on α is then the standard one from 1.8(ii): step back by a positive amount — here x itself, the natural unit — to find some $mx > \alpha - x$; step forward one term to $(m+1)x > \alpha$, and the supremum is beaten by a member of its own set. Keep the pattern: *when the hypothesis is “bounded,” the proof is “take the sup and derive a contradiction one step past it.”* It reappears at 1.21, 2.28, 2.38 and 3.14, and it is pattern 3 of the chapter’s strategy index.

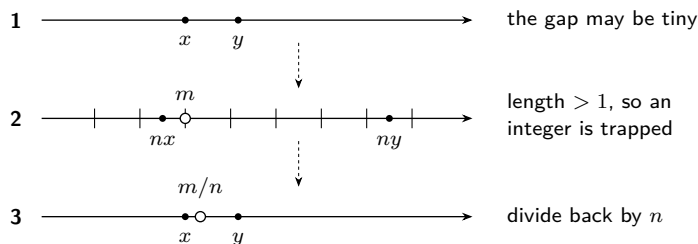
THE THREE MOVES IN PART (B)

Rudin gives the computation and not the plan. The plan is this: to fit a rational strictly between x and y , first *stretch* the gap until it is wider than 1, then *catch* an integer inside the stretched gap, then *shrink* back.

Stretch. Part (a) with the roles $x \rightsquigarrow y - x$ and $y \rightsquigarrow 1$ gives an n with $n(y - x) > 1$. The interval (nx, ny) now has length greater than 1.

Catch. An interval of length greater than 1 must contain an integer. Rudin proves this rather than asserting it, and that is what m_1, m_2 are for: they trap nx between two integers, so the set of integers exceeding nx is nonempty and bounded below, and therefore has a least element m . That gives $m - 1 \leq nx < m$. The existence of a *least* such integer is well-ordering again — the same principle used silently in 1.1. Rudin never states it as an axiom, but this step and the archimedean property both consume it.

Shrink. From $m \leq 1 + nx$ and $1 < n(y - x)$ we get $m < nx + n(y - x) = ny$, and dividing by $n > 0$ puts m/n strictly between x and y .



Stretch until an integer must fall inside, catch it, then shrink back. The integer m becomes the rational m/n between x and y .

We shall now prove the existence of n th roots of positive reals. This proof will show how the difficulty pointed out in the Introduction (irrationality of $\sqrt{2}$) can be handled in $\mathbb{R}$.

1.21 Theorem. For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$.

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof. That there is at most one such y is clear, since $0 < y_1 < y_2$ implies $y_1^n < y_2^n$.

Let E be the set consisting of all positive real numbers t such that $t^n < x$. If $t = x/(1+x)$ then $0 < t < 1$. Hence $t^n \leq t < x$. Thus $t \in E$, and E is not empty.²⁵

If $t > 1 + x$ then $t^n \geq t > x$, so that $t \notin E$. Thus $1 + x$ is an upper bound of E .

Hence Theorem 1.19 implies the existence of

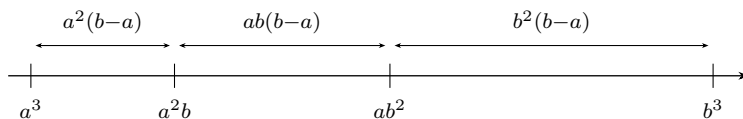
$$y = \sup E.$$

To prove that $y^n = x$ we will show that each of the inequalities $y^n < x$ and $y^n > x$ leads to a contradiction.²⁶

The identity $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1})$ yields the inequality

$$b^n - a^n < (b-a)nb^{n-1}$$

when $0 < a < b$.



The identity, drawn for $n = 3$: the walk from $a \cdot a \cdot a$ to $b \cdot b \cdot b$ replaces one factor at a time, cutting $[a^3, b^3]$ into three pieces whose widths are the three terms of the identity. Since $a < b$, every piece is shorter than the last one, whence $b^n - a^n < (b-a)nb^{n-1}$. (Drawn for $a = 1$, $b = 1.5$. For $n = 2$ this is the difference of squares; at 3.26 the same identity, with $b = 1$, $a = x$, sums the geometric series.)

Assume $y^n < x$. Choose h so that $0 < h < 1$ and

$$h < \frac{x - y^n}{n(y+1)^{n-1}}.$$

25 Two hypotheses of the LUB property are being discharged, and Rudin spends a sentence on each. This paragraph is nonemptiness; the next is boundedness. Both are required before $\sup E$ may be written down — compare the warning at 1.10.

26 Note which half of Definition 1.8 each case attacks. Assuming $y^n < x$ will contradict “ y is an upper bound”; assuming $y^n > x$ will contradict “ y is the least upper bound.” Both clauses get used exactly once, which is a reliable sign that E was chosen well.

Put $a = y$, $b = y + h$. Then

$$(y + h)^n - y^n < hn(y + h)^{n-1} < hn(y + 1)^{n-1} < x - y^n.$$

Thus $(y + h)^n < x$ and $y + h \in E$. Since $y + h > y$, this contradicts the fact that y is an upper bound of E .²⁷

Assume $y^n > x$. Put

$$k = \frac{y^n - x}{ny^{n-1}}.$$

Then $0 < k < y$. If $t \geq y - k$, we conclude that

$$y^n - t^n \leq y^n - (y - k)^n < kny^{n-1} = y^n - x.$$

Thus $t^n > x$ and $t \notin E$. It follows that $y - k$ is an upper bound of E . But $y - k < y$, which contradicts the fact that y is the least upper bound of E .

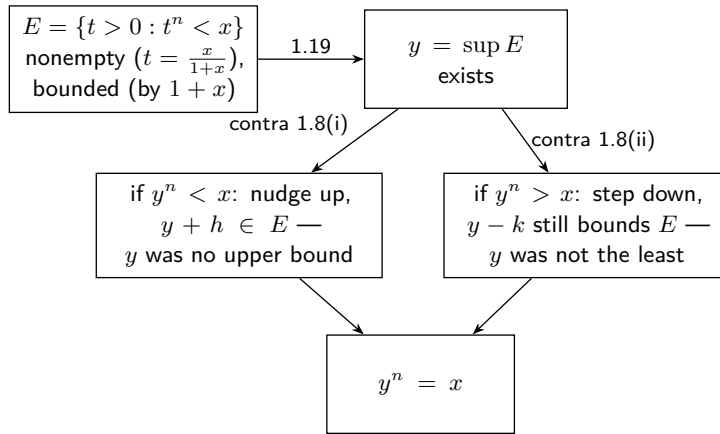
Hence $y^n = x$, and the proof is complete. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** A number with $y^n = x$ must be *produced*, and the only production tool is the least-upper-bound property — pattern 3 again. So the proof must design a set whose supremum is the root. The natural candidate: everything that undershoots, $E = \{t > 0 : t^n < x\}$.
2. **Buy the supremum.** Both tickets, each bought with one sentence: $x/(1+x)$ lies in E (nonempty), $1+x$ bounds it. Rudin's care here is the warning of 1.10 being heeded, and $y = \sup E$ exists.
3. **Set up the pincer.** By trichotomy (1.5), it suffices to kill $y^n < x$ and $y^n > x$. There is no direct computation of y^n available — y is only known as a supremum — so each case must be turned into a statement *about bounds*, the only language y speaks.
4. **Kill $y^n < x$.** If y undershoots, continuity-style estimates (the binomial identity) produce a slightly larger $y + h$ still in E — contradicting clause (i) of 1.8: y was supposed to be an upper bound. The choice of h is reverse-engineered in the next box.
5. **Kill $y^n > x$.** If y overshoots, the same estimate run downward shows $y - k$ already bounds E — contradicting clause (ii): y was supposed to be the *least* upper bound.
6. **Audit.** Clause (i) died in step 4, clause (ii) in step 5 — each spent exactly once, as in 1.11. When both halves of the supremum's definition are consumed, the set E was the right one; when a proof of this shape never uses one clause, suspect the

²⁷ Do not try to see the choice of h at a glance — nobody does. It was found backwards: write the goal $(y + h)^n < x$, bound the left side by the displayed inequality, and solve for h ; the strange side condition $h < 1$ exists only to get h out of the exponent first. The box after the proof reconstructs the whole derivation, and the k of the next paragraph is found the same way.

set.



The proof as a flow diagram: qualify E , mint $y = \sup E$, then kill both strict inequalities — each contradiction consuming a different clause of Definition 1.8. Trichotomy (1.5) does the rest.

REVERSE-ENGINEERING THE CHOICE OF H

Nobody guesses that h . It is found by writing the goal down and solving backwards, and Rudin has simply deleted the derivation.

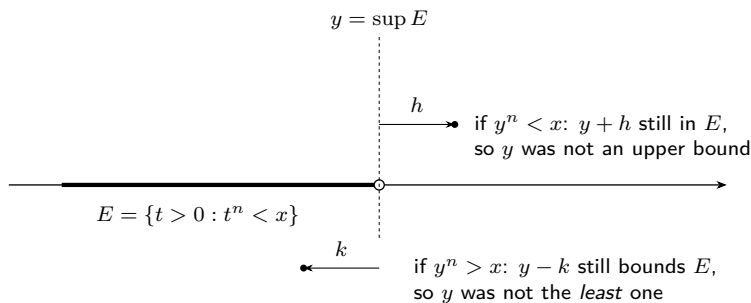
Goal. Find $h > 0$ with $(y + h)^n < x$; equivalently $(y + h)^n - y^n < x - y^n$, whose right-hand side is a fixed positive number because we assumed $y^n < x$.

Bound the left side. We cannot compute $(y + h)^n - y^n$, but the displayed inequality with $a = y$, $b = y + h$ bounds it by $hn(y + h)^{n-1}$. So it suffices to make $hn(y + h)^{n-1} \leq x - y^n$.

Get h out of the exponent. That inequality still has h buried inside $(y + h)^{n-1}$ and cannot be solved as it stands. Kill it by agreeing in advance that $h < 1$; then $(y + h)^{n-1} < (y + 1)^{n-1}$, a constant. This is the *only* reason the condition $0 < h < 1$ appears — it is bookkeeping, not mathematics.

Solve. It now suffices that $hn(y + 1)^{n-1} < x - y^n$, that is, $h < (x - y^n)/(n(y + 1)^{n-1})$, which is Rudin's line. Both conditions on h are upper bounds, so any h below the smaller of the two works, and such an h exists.

The second case is the same argument run downward, and k is found the same way; there the constraint $k < y$ plays the role that $h < 1$ played here.



Both halves of Definition 1.8 get used exactly once — a reliable sign that E was the right set to take a supremum of.

What the proof is really doing

Strip the algebra and the argument reads: $t \mapsto t^n$ is continuous, so if y^n were strictly below x we could nudge y upward and stay below x . The choice of h is a hand-rolled continuity estimate, written out because continuity is not available until 4.4 — and the machinery of Chapter 4 is not available until $\mathbb{R}$ exists, which is what we are in the middle of establishing.

Once Chapter 4 is in place the same theorem falls out of the intermediate value theorem (4.23) in three lines. Rudin proves it here anyway because he needs n th roots long before then, and because it is the first genuine demonstration that the LUB property produces numbers $\mathbb{Q}$ could not.

WHY E IS RESTRICTED TO POSITIVE T

Without positivity the uniqueness claim is simply false: for $n = 2$, $x = 4$, both 2 and -2 are roots. Rudin's “one and only one *positive* real” is doing real work, and the restriction on E is what licenses the opening line of the proof, $0 < y_1 < y_2 \Rightarrow y_1^n < y_2^n$, which requires both numbers to be positive.

Corollary. *If a and b are positive real numbers and n is a positive integer, then*

$$(ab)^{1/n} = a^{1/n}b^{1/n}.$$

Proof. Put $\alpha = a^{1/n}$, $\beta = b^{1/n}$. Then

$$ab = \alpha^n \beta^n = (\alpha\beta)^n$$

since multiplication is commutative. [Axiom (M2) in Definition 1.12.] The uniqueness assertion of Theorem 1.21 shows therefore that

$$(ab)^{1/n} = \alpha\beta = a^{1/n}b^{1/n}.$$

□

UNIQUENESS DOING THE WORK

Note what carries the argument: not a computation with roots, but the *uniqueness* half of 1.21. We exhibit one positive number, namely $\alpha\beta$, whose n th power is ab ; since 1.21 says there is only one such number, that number must be $(ab)^{1/n}$.

This is the standard way to prove identities about n th roots, and it is worth internalising because the direct approach is hopeless — there is no formula for $a^{1/n}$ to manipulate. Rudin uses the same move at 1.33(c) to get $|zw| = |z||w|$.

1.22 Decimals. We conclude this section by pointing out the relation between real numbers and decimals.

Let $x > 0$ be real. Let n_0 be the largest integer such that $n_0 \leq x$. (Note that the existence of n_0 depends on the archimedean property of $\mathbb{R}$.)²⁸ Having chosen $n_0, n_1, \dots, n_{k-1}$, let n_k denote the largest integer such that

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq x.$$

Let E be the set of these numbers

$$n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \quad (k = 0, 1, 2, \dots). \quad (1.5)$$

Then $x = \sup E$. The decimal expansion of x is

$$n_0.n_1n_2n_3\cdots. \quad (1.6)$$

Conversely, for any infinite decimal expansion (1.6) the set E of numbers (1.5) is bounded above and (1.6) is the decimal expansion of $\sup E$.

Since we shall never use decimals, we do not enter into a detailed discussion.²⁹

Why this passage is here at all

It answers the question the Introduction raised and then dropped: the sequence $1, 1.4, 1.41, \dots$ “tends to $\sqrt{2}$ ” — what does it tend to? The answer is now available and is stated in one word: $\sup$. The truncations form a set E , that set is bounded above, and 1.19 hands you its supremum. The number the decimals were approximating is that supremum, and nothing more mysterious.

Note the two directions Rudin states. Every real has a decimal expansion, and every decimal expansion names a real. Together they say the informal picture of $\mathbb{R}$ you arrived with is not wrong — it just could not be a foundation, since E having a supremum is exactly what $\mathbb{Q}$ failed to provide.

28 Rudin flags the dependence himself, which is unusual for him and a signal that it matters. Without 1.20(a) there might be no integer at all below—or above—a given x , and the decimal expansion could not even start.

29 Take that last sentence seriously — it is a warning, not modesty. Decimals are a representation of real numbers, not a definition of them, and reasoning about $\mathbb{R}$ through decimal expansions is a reliable way to go wrong (the two expansions of $1 = 0.999\dots$ are the familiar symptom). Everything in this book is proved from 1.19 and never from digits.

THE EXTENDED REAL NUMBER SYSTEM

A CONVENIENCE, NOT A NUMBER SYSTEM | One definition, no theorems. Rudin is buying a notational convenience: he wants to write

$\sup E = +\infty$ for unbounded E , and $\lim s_n = +\infty$ in Chapter 3, without special-casing every statement. Read it as an agreement about symbols.

The thing to carry away is what he pointedly does *not* do. The extended reals are given an order but not an arithmetic — “does not form a field” is stated flatly — and the conventions in (a)–(c) leave $\infty - \infty$, $0 \cdot \infty$, and ∞/∞ undefined. Those gaps are deliberate and permanent; they are why $\liminf$ and $\limsup$ in 3.16 need care, and why Theorem 3.42 has the hypotheses it does.

1.23 Definition. The *extended real number system* consists of the real field $\mathbb{R}$ and two symbols, $+\infty$ and $-\infty$. We preserve the original order in $\mathbb{R}$ and define

$$-\infty < x < +\infty$$

for every $x \in \mathbb{R}$.

It is then clear that $+\infty$ is an upper bound of every subset of the extended real number system, and that every nonempty subset has a least upper bound.³⁰ If, for example, E is a nonempty subset of real numbers which is not bounded above in $\mathbb{R}$, then $\sup E = +\infty$ in the extended real number system.

Exactly the same remarks apply to lower bounds.

The extended real number system does not form a field, but it is customary to make the following conventions:

(a) If x is real then

$$x + \infty = +\infty, \quad x - \infty = -\infty, \quad \frac{x}{+\infty} = \frac{x}{-\infty} = 0.$$

(b) If $x > 0$ then $x \cdot (+\infty) = +\infty$, $x \cdot (-\infty) = -\infty$.

(c) If $x < 0$ then $x \cdot (+\infty) = -\infty$, $x \cdot (-\infty) = +\infty$.

When it is desired to make the distinction between real numbers on the one hand and the symbols $+\infty$ and $-\infty$ on the other quite explicit, the former are called *finite*.

READ THE CONVENTIONS AS A CLOSED LIST

Every case Rudin lists involves one infinity and one *finite* x . Anything outside that pattern is simply undefined — among them $(+\infty) + (-\infty)$, $0 \cdot (\pm\infty)$, every quotient of two infinities, and every sum or product of two infinite symbols not covered above.

The omissions are deliberate, not oversights, because no assignment preserves the ordinary rules: if $\infty - \infty$ were given a value c , then adding 1 to the first term would force $c = c + 1$. (Division by zero is a separate matter — $x/0$ is undefined in any field, by (M5), and has nothing to do with the extended system.)

The practical consequence: an expression like $\sup E + \sup F$ is not auto-

³⁰ This is the payoff. Of the two hypotheses in Definition 1.10, exactly one becomes vacuous: “bounded above” is now automatic, since $+\infty$ bounds every subset. “Nonempty” is still required, and Rudin keeps it in the sentence alongside. What has been bought, at the cost of field structure, is a sup that exists for every nonempty set whether or not it is bounded.

mathematically meaningful once sup may be infinite, and theorems in Chapter 3 that add limits carry hypotheses excluding these cases. When a later proof says “provided the right-hand side is not of the form $\infty - \infty$,” this is what it is protecting against.

Why not just add ∞ and keep the field axioms?

Because it is impossible, and 1.16(b) shows why in two lines. Suppose $+\infty$ had a multiplicative inverse; convention (a) already forces $x/(+\infty) = 0$, so $1/(+\infty) = 0$, and then $1 = (+\infty) \cdot 0 = 0$ by 1.16(a) — contradicting $1 \neq 0$ in (M4).

So the extended reals are an ordered set with a partial arithmetic, and that is all they can be. Rudin’s phrasing — “it is customary to make the following conventions” — is exact: these are conventions, not theorems, and they are adopted because they make limit statements uniform, not because they are forced.

THE COMPLEX FIELD

EQUIPMENT, WITH ONE EXCEPTION | Nothing in this section is about $\mathbb{R}$, and nothing in it is hard. Rudin is stocking the shelves: Chapters 3, 7, 8 and 11 all work with complex sequences, complex series and complex-valued functions, and he wants $\mathbb{C}$ available with its arithmetic and its absolute value already proved.

The section has a definite shape. Definitions 1.24–1.29 build $\mathbb{C}$ and then recover the familiar notation $a + bi$; 1.30–1.33 introduce the conjugate and the absolute value and establish the triangle inequality; 1.34–1.35 prove the Schwarz inequality. Only two things here have downstream weight: **1.33(e)**, the triangle inequality, which is used on almost every page of Chapters 3–7, and **1.35**, the Schwarz inequality, which is the engine behind the metric on $\mathbb{R}^k$ in the next section and reappears for integrals in Exercise 6.10 and for $\mathcal{L}^2$ functions in 11.35.

Watch one point of method. Rudin defines a complex number as an *ordered pair*, verifies the field axioms by direct computation, and only then — at 1.27 — introduces i . The mysterious square root of -1 is never assumed; it is produced. That ordering is the whole reason the construction is honest.

1.24 Definition. A *complex number* is an ordered pair (a, b) of real numbers. “Ordered” means that (a, b) and (b, a) are regarded as distinct if $a \neq b$.

Let $x = (a, b)$, $y = (c, d)$ be two complex numbers. We write $x = y$ if and only if $a = c$ and $b = d$. (Note that this definition is not entirely superfluous; think of equality of rational numbers, represented

as quotients of integers.) We define

$$\begin{aligned}x + y &= (a + c, b + d) \\ xy &= (ac - bd, ad + bc).\end{aligned}$$

WHERE THE MULTIPLICATION RULE COMES FROM

The addition rule is the obvious one. The multiplication rule is not, and Rudin gives no motivation — but it is forced, not chosen. Suppose you want (a, b) to behave like $a + bi$ with $i^2 = -1$, while keeping the field axioms. Then

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i,$$

which is exactly the displayed formula. So the definition is the multiplication $\mathbb{C}$ *must* have if i is to exist at all.

Rudin declines to say this here, because saying it would presuppose i — the very thing he refuses to assume until 1.27. The formula is therefore written down bare, checked to satisfy the axioms in 1.25, and only afterwards revealed to mean what you expected. Read 1.24–1.29 as a single argument with the motivation deliberately withheld until the end.

1.25 Theorem. *These definitions of addition and multiplication turn the set of all complex numbers into a field, with $(0, 0)$ and $(1, 0)$ in the role of 0 and 1.*

Proof. We simply verify the field axioms, as listed in Definition 1.12. (Of course, we use the field structure of $\mathbb{R}$.) Let $x = (a, b)$, $y = (c, d)$, $z = (e, f)$.

(A1) is clear.

(A2) $x + y = (a + c, b + d) = (c + a, d + b) = y + x$.

(A3)

$$\begin{aligned}(x + y) + z &= (a + c, b + d) + (e, f) \\ &= (a + c + e, b + d + f) \\ &= (a, b) + (c + e, d + f) = x + (y + z).\end{aligned}$$

(A4) $x + 0 = (a, b) + (0, 0) = (a, b) = x$.

(A5) Put $-x = (-a, -b)$. Then $x + (-x) = (0, 0) = 0$.

(M1) is clear.

(M2) $xy = (ac - bd, ad + bc) = (ca - db, da + cb) = yx$.

(M3)

$$\begin{aligned}(xy)z &= (ac - bd, ad + bc)(e, f) \\ &= (ace - bde - adf - bcf, acf - bdf + ade + bce) \\ &= (a, b)(ce - df, cf + de) = x(yz).\end{aligned}$$

(M4) $1x = (1, 0)(a, b) = (a, b) = x$.

(M5) If $x \neq 0$ then $(a, b) \neq (0, 0)$, which means that at least one of the real numbers a, b is different from 0. Hence $a^2 + b^2 > 0$, by Proposition 1.18(d),³¹ and we can define

$$\frac{1}{x} = \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right).$$

Then

$$x \cdot \frac{1}{x} = (a, b) \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right) = (1, 0) = 1.$$

(D)

$$\begin{aligned} x(y + z) &= (a, b)(c + e, d + f) \\ &= (ac + ae - bd - bf, ad + af + bc + be) \\ &= (ac - bd, ad + bc) + (ae - bf, af + be) \\ &= xy + xz. \end{aligned}$$

□

Nothing here is deep, and that is the point

Every line is arithmetic in $\mathbb{R}$. That is precisely what makes the theorem worth stating: it shows the existence of $\mathbb{C}$ requires no new assumptions whatever — given $\mathbb{R}$, the complex field comes free, as a set of pairs.

Compare the situation for $\mathbb{R}$ itself, where Theorem 1.19 needed a page of Dedekind cuts. The difference is that $\mathbb{C}$ adds no *completeness*; it only adds algebra. That algebra turns out to be enormously powerful — 8.8 will show $\mathbb{C}$ is algebraically closed — but it costs nothing to obtain.

1.26 Theorem. *For any real numbers a and b we have*

$$(a, 0) + (b, 0) = (a + b, 0), \quad (a, 0)(b, 0) = (ab, 0).$$

The proof is trivial.

Theorem 1.26 shows that the complex numbers of the form $(a, 0)$ have the same arithmetic properties as the corresponding real numbers a . We can therefore identify $(a, 0)$ with a . This identification gives us the real field as a subfield of the complex field.³²

The reader may have noticed that we defined the complex numbers without any reference to the mysterious square root of -1 . We now show that the notation (a, b) is equivalent to the more customary $a + bi$.

1.27 Definition. $i = (0, 1)$.

1.28 Theorem. $i^2 = -1$.

Proof. $i^2 = (0, 1)(0, 1) = (-1, 0) = -1$. □

1.29 Theorem. *If a and b are real, then $(a, b) = a + bi$.*

31 Exactly the leverage promised at 1.18(d). Squares are nonnegative and not both a, b vanish, so $a^2 + b^2 > 0$ — and therefore the divisions below are legal. The construction of $\mathbb{C}$ depends on an order property of $\mathbb{R}$, which is worth noticing given that $\mathbb{C}$ itself cannot be ordered (Exercise 1.8).

32 The same manoeuvre as Step 9 of the appendix, where $\mathbb{Q}$ is identified with the rational cuts $\mathbb{Q}^*$. In both cases a copy of the smaller field sits inside the larger one, arithmetic and order preserved, and we agree to stop distinguishing the copy from the original. Rudin explicitly links the two identifications in Step 9; when you read the appendix, come back to this paragraph.

Proof.

$$\begin{aligned} a + bi &= (a, 0) + (b, 0)(0, 1) \\ &= (a, 0) + (0, b) = (a, b). \end{aligned}$$

□

WHAT 1.27–1.29 ACTUALLY ACCOMPLISH

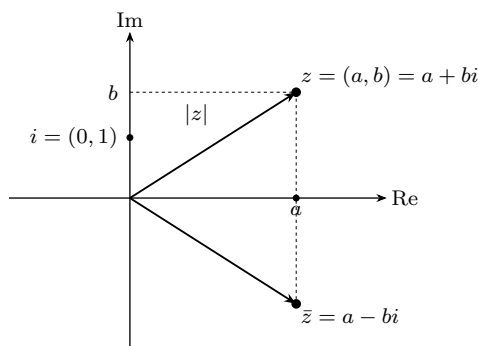
Three lines of computation, and they retire a piece of folklore. The usual story is that i is postulated — an imaginary object adjoined by fiat because we wish $x^2 + 1$ had a root. Rudin's construction assumes no such thing. $\mathbb{C}$ is built from ordered pairs of reals, its arithmetic is defined outright, and then one particular pair, $(0, 1)$, is observed to square to -1 .

So i is not an assumption; it is a *discovery* about a field that already existed. And 1.29 shows the pair notation and the $a + bi$ notation describe the same object, so you may use whichever is convenient — pairs when checking axioms, $a + bi$ when computing.

1.30 Definition. If a, b are real and $z = a + bi$, then the complex number $\bar{z} = a - bi$ is called the *conjugate* of z . The numbers a and b are the *real part* and the *imaginary part* of z , respectively.

We shall occasionally write

$$a = \operatorname{Re}(z), \quad b = \operatorname{Im}(z).$$



Conjugation is reflection in the real axis, and $|z| = \sqrt{a^2 + b^2}$ is distance from the origin. Neither is a new assumption — both are read off Definition 1.24.

1.31 Theorem. If z and w are complex, then

- (a) $\overline{z + w} = \bar{z} + \bar{w}$,
- (b) $\overline{zw} = \bar{z} \cdot \bar{w}$,
- (c) $z + \bar{z} = 2 \operatorname{Re}(z)$, $z - \bar{z} = 2i \operatorname{Im}(z)$,
- (d) $z\bar{z}$ is real and positive (except when $z = 0$).

Proof. (a), (b), and (c) are quite trivial. To prove (d), write $z = a + bi$, and note that $z\bar{z} = a^2 + b^2$. □

Conjugation is a field automorphism

Parts (a) and (b) say that $z \mapsto \bar{z}$ preserves both operations, and since $\bar{\bar{z}} = z$ it is a bijection. In the language of algebra, conjugation is an automorphism of $\mathbb{C}$ fixing $\mathbb{R}$ pointwise — and it is the only nontrivial one.

The practical use is a symmetry principle you will meet repeatedly: any polynomial identity with real coefficients that holds for z also holds for $\bar{z}$. That is why non-real roots of real polynomials come in conjugate pairs, and why 8.8 (the algebraic completeness of $\mathbb{C}$) has the consequences it does for real polynomials.

Part (d) is the one with immediate work to do: it is what makes Definition 1.32 possible, since $z\bar{z} \geq 0$ is exactly the condition needed before taking a square root by 1.21.

1.32 Definition. If z is a complex number, its *absolute value* $|z|$ is the non-negative square root of $z\bar{z}$; that is, $|z| = (z\bar{z})^{1/2}$.

The existence (and uniqueness) of $|z|$ follows from Theorem 1.21 and part (d) of Theorem 1.31.³³

Note that when x is real, then $\bar{x} = x$, hence $|x| = \sqrt{x^2}$. Thus $|x| = x$ if $x \geq 0$, $|x| = -x$ if $x < 0$.

1.33 Theorem. Let z and w be complex numbers. Then

- (a) $|z| > 0$ unless $z = 0$, $|0| = 0$,
- (b) $|\bar{z}| = |z|$,
- (c) $|zw| = |z||w|$,
- (d) $|\operatorname{Re}(z)| \leq |z|$,
- (e) $|z + w| \leq |z| + |w|$.

Proof. (a) and (b) are trivial. Put $z = a + bi$, $w = c + di$, with a, b, c, d real. Then

$$|zw|^2 = (ac - bd)^2 + (ad + bc)^2 = (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2$$

or $|zw|^2 = (|z||w|)^2$. Now (c) follows from the uniqueness assertion of Theorem 1.21.

To prove (d), note that $a^2 \leq a^2 + b^2$, hence

$$|a| = \sqrt{a^2} \leq \sqrt{a^2 + b^2}.$$

To prove (e), note that $\bar{z}w$ is the conjugate of $z\bar{w}$, so that $z\bar{w} + \bar{z}w = 2\operatorname{Re}(z\bar{w})$. Hence

$$\begin{aligned} |z + w|^2 &= (z + w)(\overline{z + w}) = z\bar{z} + z\bar{w} + \bar{z}w + w\bar{w} \\ &= |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \\ &\leq |z|^2 + 2|z\bar{w}| + |w|^2 \\ &= |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2. \end{aligned}$$

Now (e) follows by taking square roots. $\square$

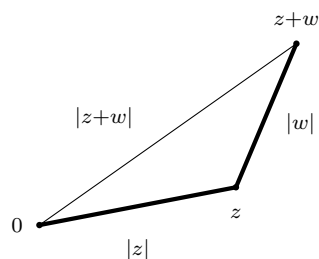
33 Both citations are load-bearing, and note the gap they leave. For $z \neq 0$, 1.31(d) makes $z\bar{z}$ a positive real and 1.21 — which requires a positive input — returns its unique positive square root. For $z = 0$ we have $z\bar{z} = 0$, which 1.21 does not cover; that case is settled separately by the convention $|0| = 0$, recorded in 1.33(a). This is the first place the work of §1.21 gets spent.

THE STANDARD PROOF OF A TRIANGLE INEQUALITY

The shape of this argument recurs throughout analysis, and it is worth extracting because you will need to reproduce it: to bound $|z + w|$, do not work with $|z + w|$ — work with $|z + w|^2$, expand it as $(z + w)(\overline{z + w})$, bound the cross-term, and take square roots at the end.

The cross-term is the only real step, and it goes in two moves: 1.31(c) turns $z\bar{w} + \bar{z}w$ into $2\operatorname{Re}(z\bar{w})$, then (d) bounds Re by modulus and (c) factors it as $|z||w|$.

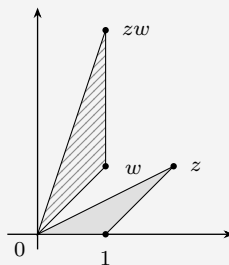
The identical strategy proves 1.37(e) for $\mathbb{R}^k$ in the next section, with the Schwarz inequality playing the part of (d). Recognising the pattern once saves reading the second proof.



Part (e) is about an actual triangle: the straight route from 0 to $z + w$ never beats the detour through z . Equality is the degenerate triangle — z and w pointing the same way, which is exactly $\operatorname{Re}(z\bar{w}) = |z\bar{w}|$, the one inequality in the proof.

Multiplication rotates and scales — (c) drawn instead of computed

Part (c) also has a picture. Multiplying every point of the plane by w rotates it by the angle of w and stretches it by the factor $|w|$: the triangle with vertices $0, w, zw$ is the triangle $0, 1, z$ so rotated and stretched.



The shaded triangle $0, 1, z$ and the hatched triangle $0, w, zw$ are similar, with ratio $|w|$. (Drawn for $z = 2 + i$, $w = 1 + i$; the similarity is general.)

Part (c) is the stretch factor read off the picture: the side $0 \rightarrow zw$ is the side $0 \rightarrow z$ magnified by $|w|$, so $|zw| = |z||w|$. Rudin proves it by the $z\bar{z}$ computation instead, because the rotation–scaling description presupposes angles — sines and cosines — which do not officially exist until Chapter 8. The picture is why the computation was never in doubt; the computation is what makes the picture legal. Taking $w = \bar{z}$ folds the rotation flat onto the real axis: $z\bar{z} = |z|^2$ is

||| this figure with the hatched triangle lying on its back.

1.34 Notation. If $x_1, \dots, x_n$ are complex numbers, we write

$$x_1 + x_2 + \cdots + x_n = \sum_{j=1}^n x_j.$$

We conclude this section with an important inequality, usually known as the *Schwarz inequality*.

1.35 Theorem. If $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are complex numbers, then

$$\left| \sum_{j=1}^n a_j \bar{b}_j \right|^2 \leq \sum_{j=1}^n |a_j|^2 \sum_{j=1}^n |b_j|^2.$$

Proof. Put $A = \sum |a_j|^2$, $B = \sum |b_j|^2$, $C = \sum a_j \bar{b}_j$ (in all sums in this proof, j runs over the values $1, \dots, n$). If $B = 0$, then $b_1 = \cdots = b_n = 0$, and the conclusion is trivial. Assume therefore that $B > 0$. By Theorem 1.31 we have

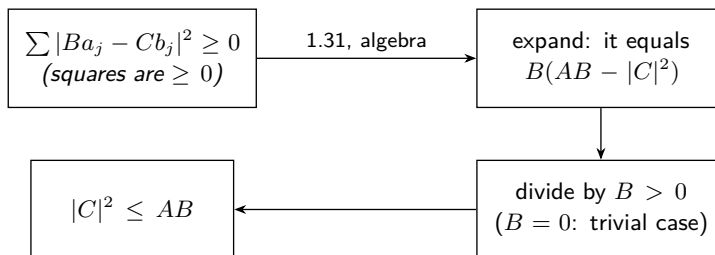
$$\begin{aligned} \sum |Ba_j - Cb_j|^2 &= \sum (Ba_j - Cb_j)(\overline{Ba_j - Cb_j}) \\ &= B^2 \sum |a_j|^2 - B\bar{C} \sum a_j \bar{b}_j \\ &\quad - BC \sum \bar{a}_j b_j + |C|^2 \sum |b_j|^2 \\ &= B^2 A - B|C|^2 \\ &= B(AB - |C|^2). \end{aligned}$$

Since each term in the first sum is nonnegative, we see that

$$B(AB - |C|^2) \geq 0.$$

Since $B > 0$, it follows that $AB - |C|^2 \geq 0$. This is the desired inequality.

□



The proof as a flow diagram: one nonnegative square, expanded and collapsed. All the invention is in choosing which square — the box below recovers that choice.

WHERE $BA_J - CB_J$ COMES FROM

This is the second rabbit of the chapter, after the h of 1.21, and like that one it is reverse-engineered rather than guessed.

The only fact available is that a sum of squared moduli is nonnegative. So the proof must begin with $\sum |\cdot|^2 \geq 0$ for some cleverly chosen expression, and end with the desired inequality. Work backwards: we want $AB - |C|^2 \geq 0$, so we want the sum to evaluate to a positive multiple of $AB - |C|^2$.

Try $\sum |\lambda a_j - \mu b_j|^2$ with constants to be chosen. Expanding gives

$$|\lambda|^2 A - \lambda \bar{\mu} \bar{C} - \bar{\lambda} \mu C + |\mu|^2 B.$$

Choosing $\lambda = B$ (real) and $\mu = C$ makes the middle terms $-B\bar{C}C - BC\bar{C} = -2B|C|^2$ and the last term $|C|^2 B$, so the whole collapses to $B^2 A - B|C|^2 = B(AB - |C|^2)$. The factor B is then stripped off using $B > 0$.

So the choice is not inspired — it is what you get by writing down the general quadratic and solving for the constants that make it collapse. The hypothesis $B > 0$ and the separate treatment of $B = 0$ exist purely because the division at the last step needs it.

Equality, and what the inequality means

Equality holds precisely when (a_j) and (b_j) are *linearly dependent*. That is Exercise 1.15, and the proof above shows why — with a caveat about how it is usually stated. When $B > 0$, equality forces $\sum |Ba_j - Cb_j|^2 = 0$, and a sum of nonnegative terms vanishes only if every term does, giving $Ba_j = Cb_j$ for all j : the two vectors are proportional. When $B = 0$ every b_j vanishes, both sides of the inequality are 0, and equality holds trivially — but “proportional” now says nothing, since $Ba_j = Cb_j$ reduces to $0 = 0$. Linear dependence is the formulation that covers both cases.

Geometrically, once 1.36 defines the inner product on $\mathbb{R}^k$, this says $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}||\mathbf{y}|$, which is the statement that $\cos \theta$ lies between -1 and 1 . Rudin proves the inequality first and lets the geometry follow, rather than the reverse; the inequality is the substance and the picture is the illustration.

Its reach is long. It gives the triangle inequality in $\mathbb{R}^k$ at 1.37(e), hence the metric that all of Chapter 2 is built on; it reappears for integrals in Exercise 6.10 — Hölder’s inequality, of which Rudin there remarks “Theorem 1.35 is a very special case” — and for $\mathcal{L}^2$ functions at 11.35, where it is the same statement about a different inner product.

EUCLIDEAN SPACES

THE LAST PIECE OF EQUIPMENT | Three items, and the section exists for one sentence: Remark 1.38, which observes that $\mathbb{R}^k$ is a metric space. Everything in Chapter 2 — open sets, compactness, connectedness — is developed for metric spaces, and $\mathbb{R}^k$ is the example that motivates all of it. This section supplies the distance function and verifies the three properties Chapter 2 will demand.

Note what is *not* claimed. Rudin gives $\mathbb{R}^k$ a vector space structure and an inner product, but no multiplication of vectors by vectors — $\mathbb{R}^k$ is not a field for $k > 1$. The only exception is $k = 2$, which coincides with $\mathbb{C}$; 1.38 remarks on the coincidence, and it is worth keeping straight that $\mathbb{R}^2$ and $\mathbb{C}$ are the same metric space but only $\mathbb{C}$ has a product.

1.36 Definitions. For each positive integer k , let $\mathbb{R}^k$ be the set of all ordered k -tuples

$$\mathbf{x} = (x_1, x_2, \dots, x_k),$$

where $x_1, \dots, x_k$ are real numbers, called the *coordinates* of $\mathbf{x}$. The elements of $\mathbb{R}^k$ are called *points*, or *vectors*, especially when $k > 1$. We shall denote vectors by boldfaced letters. If $\mathbf{y} = (y_1, \dots, y_k)$ and if α is a real number, put

$$\begin{aligned}\mathbf{x} + \mathbf{y} &= (x_1 + y_1, \dots, x_k + y_k) \\ \alpha \mathbf{x} &= (\alpha x_1, \dots, \alpha x_k)\end{aligned}$$

so that $\mathbf{x} + \mathbf{y} \in \mathbb{R}^k$ and $\alpha \mathbf{x} \in \mathbb{R}^k$. This defines addition of vectors, as well as multiplication of a vector by a real number (a *scalar*). These two operations satisfy the commutative, associative, and distributive laws (the proof is trivial, in view of the analogous laws for the real numbers) and make $\mathbb{R}^k$ into a *vector space over the real field*. The zero element of $\mathbb{R}^k$ (sometimes called the *origin* or the *null vector*) is the point $\mathbf{0}$, all of whose coordinates are 0.

We also define the so-called “inner product” (or scalar product) of $\mathbf{x}$ and $\mathbf{y}$ by

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^k x_i y_i$$

and the *norm* of $\mathbf{x}$ by

$$|\mathbf{x}| = (\mathbf{x} \cdot \mathbf{x})^{1/2} = \left(\sum_{i=1}^k x_i^2 \right)^{1/2}.$$

The structure now defined (the vector space $\mathbb{R}^k$ with the above inner product and norm) is called *euclidean k -space*.³⁴

1.37 Theorem. Suppose $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^k$, and α is real. Then

34 Three separate structures are being stacked here, and later chapters use them independently: a vector space (addition and scaling, used throughout Chapter 9), an inner product (used for orthogonality and in Chapter 11), and a norm (used to make a metric, which is all Chapter 2 needs). The norm is derived from the inner product; the inner product is not derived from anything.

- (a) $|\mathbf{x}| \geq 0$;
- (b) $|\mathbf{x}| = 0$ if and only if $\mathbf{x} = \mathbf{0}$;
- (c) $|\alpha\mathbf{x}| = |\alpha| |\mathbf{x}|$;
- (d) $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| |\mathbf{y}|$;
- (e) $|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$;
- (f) $|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$.

Proof. (a), (b), and (c) are obvious, and (d) is an immediate consequence of the Schwarz inequality.³⁵ By (d) we have

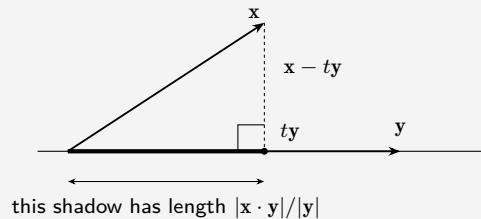
$$\begin{aligned}
 |\mathbf{x} + \mathbf{y}|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) \\
 &= \mathbf{x} \cdot \mathbf{x} + 2\mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{y} \\
 &\leq |\mathbf{x}|^2 + 2|\mathbf{x}| |\mathbf{y}| + |\mathbf{y}|^2 \\
 &= (|\mathbf{x}| + |\mathbf{y}|)^2,
 \end{aligned}$$

so that (e) is proved. Finally (f) follows from (e) if we replace $\mathbf{x}$ by $\mathbf{x} - \mathbf{y}$ and $\mathbf{y}$ by $\mathbf{y} - \mathbf{z}$. $\square$

The geometry of 1.37(d) and (e)

Parts (d) and (e) are the two facts of Euclidean geometry that survive into arbitrary dimension, and each has a picture that is also its proof.

(d) Schwarz is a statement about projection. Assume $\mathbf{y} \neq \mathbf{0}$ — otherwise both sides of (d) vanish and there is nothing to prove. Drop a perpendicular from $\mathbf{x}$ onto the line through $\mathbf{y}$. The foot sits at $t\mathbf{y}$ with $t = (\mathbf{x} \cdot \mathbf{y})/|\mathbf{y}|^2$, and the leftover piece $\mathbf{x} - t\mathbf{y}$ is perpendicular to $\mathbf{y}$.



The shadow of $\mathbf{x}$ on $\mathbf{y}$ cannot be longer than $\mathbf{x}$ itself.

Now compute the length of that leftover piece:

$$0 \leq |\mathbf{x} - t\mathbf{y}|^2 = |\mathbf{x}|^2 - \frac{(\mathbf{x} \cdot \mathbf{y})^2}{|\mathbf{y}|^2},$$

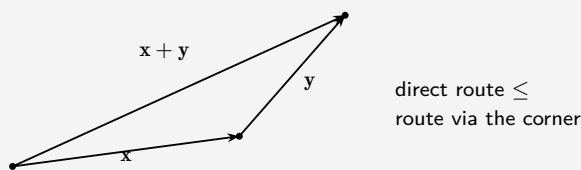
and rearranging gives exactly $(\mathbf{x} \cdot \mathbf{y})^2 \leq |\mathbf{x}|^2 |\mathbf{y}|^2$. That is Schwarz, and no angle was needed — only that a squared length is not negative. Equality holds precisely when the leftover piece vanishes, that is when $\mathbf{x}$ lies on the line through $\mathbf{y}$: the linear dependence of Exercise 1.15.

35 Take 1.35 with all a_j, b_j real; conjugation does nothing, and the statement becomes $(\sum x_i y_i)^2 \leq \sum x_i^2 \sum y_i^2$, which is (d) squared. This is the only place the previous section's main theorem is used, and it is the reason that section had to come first.

This is also Rudin's proof of 1.35 in disguise. His $\sum |Ba_j - Cb_j|^2 \geq 0$ is the same "a squared length is not negative," with $Ba_j - Cb_j$ playing the part of the leftover piece — and his separate treatment of $B = 0$ is exactly the $\mathbf{y} = \mathbf{0}$ case set aside above.

A third way to say it, if you prefer algebra to pictures: the function $q(t) = |\mathbf{x} - t\mathbf{y}|^2$ is a quadratic in t that is never negative, so its discriminant cannot be positive, and $4(\mathbf{x} \cdot \mathbf{y})^2 - 4|\mathbf{y}|^2|\mathbf{x}|^2 \leq 0$. The three proofs are one proof: the projection is where q attains its minimum, and the discriminant condition says that minimum is not below zero.

(e) **The triangle inequality is a triangle.** Lay $\mathbf{y}$ head to tail after $\mathbf{x}$; the direct route to $\mathbf{x} + \mathbf{y}$ is no longer than going via the corner.



Part (f) is the same picture read as three points: going from $\mathbf{x}$ to $\mathbf{z}$ directly beats detouring through $\mathbf{y}$.

But here the picture is not a proof, and it is worth being clear why. For Schwarz the figure could be turned into an argument, because every step was an identity about inner products. Not so here. "The straight path is shortest" is a theorem of Euclidean geometry, and in Rudin's development Euclidean geometry does not yet exist — 1.36 has just *defined* distance by a formula, and 1.37 is where the familiar properties of that formula are established for the first time. Appealing to the picture would assume what is being proved. The figure shows you what 1.37(e) means; the algebra, via Schwarz, is what makes it true.

PART (F) IS THE TRIANGLE INEQUALITY YOU WILL ACTUALLY USE

Parts (e) and (f) say the same thing twice, and it is worth seeing why both are stated. Part (e) is about *vectors*: the length of a sum is at most the sum of the lengths. Part (f) is about *points*: the distance from $\mathbf{x}$ to $\mathbf{z}$ is at most the distance via any detour through $\mathbf{y}$.

The substitution converting one into the other is worth doing once by hand. Setting $\mathbf{u} = \mathbf{x} - \mathbf{y}$ and $\mathbf{v} = \mathbf{y} - \mathbf{z}$, we have $\mathbf{u} + \mathbf{v} = \mathbf{x} - \mathbf{z}$, so (e) reads $|\mathbf{x} - \mathbf{z}| \leq |\mathbf{x} - \mathbf{y}| + |\mathbf{y} - \mathbf{z}|$.

Form (f) is the one that generalises. When Chapter 2 defines a metric space there is no addition available, so vectors cannot be summed and (e) has no meaning — but distances between three

points still make sense, and (f) becomes axiom (c) of Definition 2.15.

[36] This is the entire purpose of the section, stated in one sentence and easy to read past. Compare the three parts he names with Definition 2.15: (a) and (b) give $d(x, y) > 0$ for $x \neq y$ and $d(x, x) = 0$; (f) is the triangle inequality. Symmetry, the remaining axiom, follows from (c) with $\alpha = -1$. Rudin has just verified, in advance, that $\mathbb{R}^k$ is an example of the structure Chapter 2 will study abstractly.

1.38 Remarks. Theorem 1.37(a), (b), and (f) will allow us (see Chap. 2) to regard $\mathbb{R}^k$ as a *metric space*.³⁶

$\mathbb{R}^1$ (the set of all real numbers) is usually called *the line*, or *the real line*. Likewise, $\mathbb{R}^2$ is called *the plane*, or *the complex plane* (compare Definitions 1.24 and 1.36). In these two cases, the norm is just the absolute value of the corresponding real or complex number.

$\mathbb{R}^2$ AND $\mathbb{C}$ ARE THE SAME SPACE BUT NOT THE SAME OBJECT

As metric spaces they are identical: $(a, b) \in \mathbb{R}^2$ and $a + bi \in \mathbb{C}$ have the same norm $\sqrt{a^2 + b^2}$, so every statement about convergence, open sets or compactness transfers word for word.

They differ in algebra. $\mathbb{C}$ has a multiplication making it a field (1.25); $\mathbb{R}^2$ has only the inner product, which produces a *scalar*, not a vector. So zw makes sense for $z, w \in \mathbb{C}$ and $\mathbf{x}\mathbf{y}$ does not for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$.

The distinction matters when a theorem is stated for one and you want it for the other. Results resting only on the metric — most of Chapter 2, and the convergence theorems of Chapter 3 — carry across freely. Results using multiplication or division, such as the whole theory of power series in Chapter 8, belong to $\mathbb{C}$ alone.

THE CHAPTER PROPER, IN ONE PARAGRAPH

$\mathbb{Q}$ is unfit for analysis because a bounded set can fail to have a least upper bound (1.1). Name that failure (1.8–1.10), develop just enough algebra to state what $\mathbb{R}$ must be (1.12–1.18), assert a field free of the failure (1.19), and the consequences follow at once: infima exist too (1.11), $\mathbb{N}$ is unbounded and $\mathbb{Q}$ is dense (1.20), and every positive real has every root (1.21). The remaining sections build the objects later chapters will work in — the extended reals for unbounded suprema, $\mathbb{C}$ for Chapters 3 and 8, $\mathbb{R}^k$ for Chapter 2 onward — and prove one inequality of lasting importance, Schwarz’s (1.35). Nothing in the rest of the book uses more about $\mathbb{R}$ than 1.19. When a later proof says “by completeness,” that is what it means.

APPENDIX

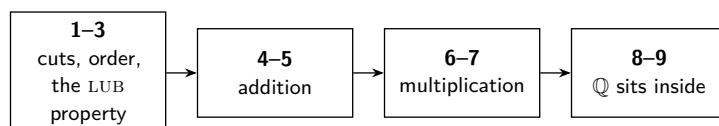
WHAT THE APPENDIX IS DOING, AND WHEN TO READ IT

This is the promissory note from 1.19 being paid. Rudin asserted that an ordered field with the least-upper-bound property exists; here he builds one out of $\mathbb{Q}$ and nothing else.

You may skip it. Nothing in Chapters 2–11 depends on having read it, and Rudin says in the Preface it may be “studied and enjoyed whenever the time seems ripe.” What it buys you is the knowledge that 1.19 is a theorem rather than an article of faith. He is not alone in deferring it: Yu Pin’s Tsinghua notes call real-number theory “very important for the completeness of the knowledge system, but not indispensable for understanding the basic ideas of analysis,” and the USTC lecture notes postpone it to a third volume.

The idea in one sentence. A real number is identified with the set of all rationals below it. That set — a *cut* — is a *subset* of $\mathbb{Q}$, so no new kind of object has to be conjured up; $\sqrt{2}$ becomes $\{p \in \mathbb{Q} : p < 0 \text{ or } p^2 < 2\}$, a perfectly ordinary collection of rationals which exists whether or not $\sqrt{2}$ does.

The nine steps, and which ones matter.



Step 3 is the whole point — it is where the least-upper-bound property appears, and its proof is four lines: the supremum of a set of cuts is their *union*. That is the payoff of representing numbers as sets. Steps 4–7 are bookkeeping, verifying field axioms one at a time, and Rudin openly omits most of Step 6 as too similar to Step 4. Steps 8–9 establish the second sentence of 1.19, that $\mathbb{Q}$ is a subfield.

If you read only two things, read Step 1 (what a cut is, and why property (III) is there) and Step 3.

Theorem 1.19 will be proved in this appendix by constructing $\mathbb{R}$ from $\mathbb{Q}$. We shall divide the construction into several steps.

Step 1 The members of $\mathbb{R}$ will be certain subsets of $\mathbb{Q}$, called *cuts*. A cut is, by definition, any set $\alpha \subseteq \mathbb{Q}$ with the following three properties.

- (I) α is not empty, and $\alpha \neq \mathbb{Q}$.
- (II) If $p \in \alpha$, $q \in \mathbb{Q}$, and $q < p$, then $q \in \alpha$.
- (III) If $p \in \alpha$, then $p < r$ for some $r \in \alpha$.

The letters $p, q, r, \dots$ will always denote rational numbers, and $\alpha, \beta, \gamma, \dots$ will denote cuts.

Note that (III) simply says that α has no largest member; (II) implies two facts which will be used freely:

If $p \in \alpha$ and $q \notin \alpha$, then $p < q$.

If $r \notin \alpha$ and $r < s$, then $s \notin \alpha$.

WHAT THE THREE PROPERTIES ARE FOR

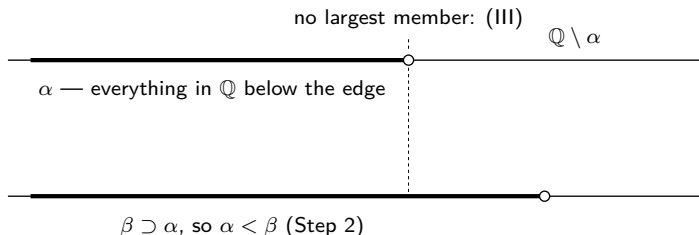
Picture the cut α as “everything strictly below some point on the line.” Each condition rules out a degenerate set that would spoil that picture.

(I) excludes the two extremes. $\emptyset$ and $\mathbb{Q}$ would correspond to $-\infty$ and $+\infty$, which are not real numbers.

(II) says α is downward closed — it is an initial segment of $\mathbb{Q}$, not an arbitrary subset. Without it $\{0, 5\}$ would qualify, and nothing about order would work.

(III) says α has no largest member, which forces the cut to be the set of rationals *strictly* below its edge. This is the delicate one, and its purpose is uniqueness: without it, the point 2 would be represented by two different cuts, $\{p < 2\}$ and $\{p \leq 2\}$, and the map from cuts to numbers would be two-to-one at every rational. Exercise 1.20 asks what breaks if you drop it — the answer is that the additive inverse (A5) fails.

The two consequences of (II) that Rudin records are the workhorses of the whole appendix. Both say the same thing — a cut has no gaps and no stragglers — and he uses them dozens of times without further comment.



A cut is a leftward ray of $\mathbb{Q}$ with no endpoint. Order is containment; one cut is smaller when it is a proper subset.

Step 2 Define “ $\alpha < \beta$ ” to mean: α is a proper subset of β .

Let us check that this meets the requirements of Definition 1.5.

If $\alpha < \beta$ and $\beta < \gamma$ it is clear that $\alpha < \gamma$. (A proper subset of a proper subset is a proper subset.) It is also clear that at most one of the three relations

$$\alpha < \beta, \quad \alpha = \beta, \quad \beta < \alpha$$

can hold for any pair α, β . To show that at least one holds, assume that the first two fail. Then α is not a subset of β . Hence there is a $p \in \alpha$ with $p \notin \beta$. If $q \in \beta$, it follows that $q < p$ (since $p \notin \beta$), hence $q \in \alpha$, by (II). Thus $\beta \subset \alpha$. Since $\beta \neq \alpha$, we conclude $\beta < \alpha$.³⁷

Thus $\mathbb{R}$ is now an ordered set.

Step 3 The ordered set $\mathbb{R}$ has the least upper bound property.

³⁷ Trichotomy is the only part needing work, because containment of arbitrary sets is a partial order — two sets can be incomparable. What rules that out here is (II): downward-closed subsets of a totally ordered set are automatically nested. This is the first place property (II) earns its keep.

To prove this, let A be a nonempty subset of $\mathbb{R}$, and assume that $\beta \in \mathbb{R}$ is an upper bound of A . Define γ to be the union of all $\alpha \in A$. In other words, $p \in \gamma$ if and only if $p \in \alpha$ for some $\alpha \in A$. We shall prove that $\gamma \in \mathbb{R}$ and that $\gamma = \sup A$.

Since A is not empty, there exists an $\alpha_0 \in A$. This α_0 is not empty. Since $\alpha_0 \subset \gamma$, γ is not empty. Next, $\gamma \subset \beta$ (since $\alpha \subset \beta$ for every $\alpha \in A$), and therefore $\gamma \neq \mathbb{Q}$. Thus γ satisfies property (I). To prove (II) and (III), pick $p \in \gamma$. Then $p \in \alpha_1$ for some $\alpha_1 \in A$. If $q < p$, then $q \in \alpha_1$, hence $q \in \gamma$; this proves (II). If $r \in \alpha_1$ is so chosen that $r > p$, we see that $r \in \gamma$ (since $\alpha_1 \subset \gamma$), and therefore γ satisfies (III). Thus $\gamma \in \mathbb{R}$.

It is clear that $\alpha \leq \gamma$ for every $\alpha \in A$.

Suppose $\delta < \gamma$. Then there is an $s \in \gamma$ such that $s \notin \delta$. Since $s \in \gamma$, $s \in \alpha$ for some $\alpha \in A$. Hence $\delta < \alpha$ and δ is not an upper bound of A .

This gives the desired result: $\gamma = \sup A$.

Why both hypotheses of 1.10 appear, and where

Watch where each is consumed. *Nonempty* gives an $\alpha_0 \in A$, which is what makes γ nonempty — half of property (I). *Bounded above* gives a β containing every α , hence $\gamma \subset \beta \neq \mathbb{Q}$ — the other half.

Without either, the union might not be a cut: an empty union is $\emptyset$, and an unbounded union is all of $\mathbb{Q}$. Both are excluded by (I) precisely because they are the cuts that would represent $-\infty$ and $+\infty$. This is the same observation as the warning at 1.10, now visible from the inside: the two hypotheses are not conveniences, they are what keeps the answer inside the system.

The last two paragraphs are the two clauses of Definition 1.8: γ is an upper bound, and nothing smaller is. Note how cheap the second is — if δ is properly contained in the union, it must miss an element that some α contains, so α already exceeds it.

Step 4 If $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$, we define $\alpha + \beta$ to be the set of all sums $r + s$, where $r \in \alpha$ and $s \in \beta$.

We define 0^* to be the set of all negative rational numbers. It is clear that 0^* is a cut. We verify that *the axioms for addition* (see Definition 1.12) *hold in $\mathbb{R}$, with 0^* playing the role of 0.*

(A1) We have to show that $\alpha + \beta$ is a cut. It is clear that $\alpha + \beta$ is a nonempty subset of $\mathbb{Q}$. Take $r' \notin \alpha$, $s' \notin \beta$. Then $r' + s' > r + s$ for all choices of $r \in \alpha$, $s \in \beta$. Thus $r' + s' \notin \alpha + \beta$. It follows that $\alpha + \beta$ has property (I).

Pick $p \in \alpha + \beta$. Then $p = r + s$, with $r \in \alpha$, $s \in \beta$. If $q < p$, then $q - s < r$, so $q - s \in \alpha$, and $q = (q - s) + s \in \alpha + \beta$. Thus (II) holds. Choose $t \in \alpha$ so that $t > r$. Then $p < t + s$ and $t + s \in \alpha + \beta$. Thus (III) holds.

(A2) $\alpha + \beta$ is the set of all $r + s$, with $r \in \alpha$, $s \in \beta$. By the same

definition, $\beta + \alpha$ is the set of all $s + r$. Since $r + s = s + r$ for all $r \in \mathbb{Q}$, $s \in \mathbb{Q}$, we have $\alpha + \beta = \beta + \alpha$.

(A3) As above, this follows from the associative law in $\mathbb{Q}$.

(A4) If $r \in \alpha$ and $s \in 0^*$, then $r + s < r$, hence $r + s \in \alpha$. Thus $\alpha + 0^* \subset \alpha$. To obtain the opposite inclusion, pick $p \in \alpha$, and pick $r \in \alpha$, $r > p$. Then $p - r \in 0^*$, and $p = r + (p - r) \in \alpha + 0^*$. Thus $\alpha \subset \alpha + 0^*$. We conclude that $\alpha + 0^* = \alpha$.³⁸

(A5) Fix $\alpha \in \mathbb{R}$. Let β be the set of all p with the following property:

There exists $r > 0$ such that $-p - r \notin \alpha$.

In other words, some rational number smaller than $-p$ fails to be in α .

We show that $\beta \in \mathbb{R}$ and that $\alpha + \beta = 0^*$.

If $s \notin \alpha$ and $p = -s - 1$, then $-p - 1 \notin \alpha$, hence $p \in \beta$. So β is not empty. If $q \in \alpha$, then $-q \notin \beta$. So $\beta \neq \mathbb{Q}$. Hence β satisfies (I).

Pick $p \in \beta$ and $r > 0$ so that $-p - r \notin \alpha$. If $q < p$, then $-q - r > -p - r$, hence $-q - r \notin \alpha$. Thus $q \in \beta$ and (II) holds. Put $t = p + (r/2)$. Then $t > p$ and $-t - (r/2) = -p - r \notin \alpha$, so that $t \in \beta$. Hence β satisfies (III).

We have proved that $\beta \in \mathbb{R}$.

If $r \in \alpha$ and $s \in \beta$, then $-s \notin \alpha$, hence $r < -s$, $r + s < 0$. Thus $\alpha + \beta \subset 0^*$.

To prove the opposite inclusion, pick $v \in 0^*$, put $w = -v/2$. Then $w > 0$ and there is an integer n such that $nw \in \alpha$ but $(n+1)w \notin \alpha$. (Note that this depends on the fact that $\mathbb{Q}$ has the archimedean property!) Put $p = -(n+2)w$. Then $p \in \beta$, since $-p - w \notin \alpha$, and

$$v = nw + p \in \alpha + \beta.$$

Thus $0^* \subset \alpha + \beta$.

We conclude that $\alpha + \beta = 0^*$. This β will of course be denoted by $-\alpha$.

WHY $-\alpha$ IS DEFINED SO AWKWARDLY

The natural guess is that $-\alpha$ should be $\{-q : q \notin \alpha\}$ — reflect the complement. That is almost right and fails on exactly one point. If α is the cut of a rational r , then $\mathbb{Q} \setminus \alpha$ has a smallest element, namely r itself. Reflecting gives a set with a *largest* element $-r$, violating property (III). So the naive definition does not always produce a cut.

Rudin's “there exists $r > 0$ with $-p - r \notin \alpha$ ” repairs this by de-

38 The reverse inclusion is where property (III) is used, and it is used essentially: we need some $r \in \alpha$ strictly above p so that $p - r$ is negative. If α had a largest member p , no such r would exist and $\alpha + 0^*$ would be missing p .

manding strict room to spare: p qualifies only if something *strictly below* $-p$ already lies outside α . That extra r shaves off the single offending endpoint and nothing else. You can see the repair working in the verification of (III), where $t = p + r/2$ is produced by spending half the slack.

The archimedean appeal at the end is the other subtlety, and Rudin flags it himself with an exclamation mark. To hit an arbitrary $v \in 0^*$ we need integer multiples of w to eventually escape α ; that is exactly 1.20(a), holding in $\mathbb{Q}$. Note the logical order — the archimedean property of $\mathbb{Q}$ is elementary and assumed, and it is being used to build $\mathbb{R}$, not derived from it.

Step 5 Having proved that the addition defined in Step 4 satisfies axioms (A) of Definition 1.12, it follows that Proposition 1.14 is valid in $\mathbb{R}$, and we can prove one of the requirements in Definition 1.17:

If $\alpha, \beta, \gamma \in \mathbb{R}$ and $\beta < \gamma$, then $\alpha + \beta < \alpha + \gamma$.

Indeed, it is obvious from the definition of $+$ in $\mathbb{R}$ that $\alpha + \beta \subset \alpha + \gamma$; if we had $\alpha + \beta = \alpha + \gamma$, the cancellation law (Proposition 1.14) would imply $\beta = \gamma$.³⁹

It also follows that $\alpha > 0^*$ if and only if $-\alpha < 0^*$.

Step 6 Multiplication is a little more bothersome than addition in the present context, since products of negative rationals are positive. For this reason we confine ourselves first to $\mathbb{R}^+$, the set of all $\alpha \in \mathbb{R}$ with $\alpha > 0^*$.

If $\alpha \in \mathbb{R}^+$ and $\beta \in \mathbb{R}^+$, we define $\alpha\beta$ to be the set of all p such that $p \leq rs$ for some choice of $r \in \alpha$, $s \in \beta$, $r > 0$, $s > 0$.⁴⁰

We define 1^* to be the set of all $q < 1$.

Then *the axioms (M) and (D) of Definition 1.12 hold, with $\mathbb{R}^+$ in place of F , and with 1^* in the role of 1 .*

The proofs are so similar to the ones given in detail in Step 4 that we omit them.

Note, in particular, that the second requirement of Definition 1.17 holds: If $\alpha > 0^*$ and $\beta > 0^*$, then $\alpha\beta > 0^*$.

Step 7 We complete the definition of multiplication by setting $\alpha 0^* = 0^* \alpha = 0^*$, and by setting

$$\alpha\beta = \begin{cases} (-\alpha)(-\beta) & \text{if } \alpha < 0^*, \beta < 0^*, \\ -[(-\alpha)\beta] & \text{if } \alpha < 0^*, \beta > 0^*, \\ -[\alpha(-\beta)] & \text{if } \alpha > 0^*, \beta < 0^*. \end{cases}$$

The products on the right were defined in Step 6.

Having proved (in Step 6) that the axioms (M) hold in $\mathbb{R}^+$, it is now perfectly simple to prove them in $\mathbb{R}$, by repeated application of the identity $\gamma = -(-\gamma)$, which is part of Proposition 1.14. (See Step 5.)

[39] Exactly the dividend promised at 1.13(c). Proposition 1.14 was proved once, for an arbitrary set satisfying axioms (A); having checked those axioms in Step 4, Rudin may now invoke cancellation for cuts without reproving anything. The one-time purchase is being collected.

[40] The definition is not “the set of all products rs ” — that would fail property (II), since the products of positive rationals omit everything negative. The clause $p \leq rs$ patches this by throwing in everything below. Contrast Step 4, where the sums $r + s$ already sweep out a downward-closed set on their own.

The proof of the distributive law

$$\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$$

breaks into cases. For instance, suppose $\alpha > 0^*$, $\beta < 0^*$, $\beta + \gamma > 0^*$. Then $\gamma = (\beta + \gamma) + (-\beta)$, and (since we already know that the distributive law holds in $\mathbb{R}^+$)

$$\alpha\gamma = \alpha(\beta + \gamma) + \alpha \cdot (-\beta).$$

But $\alpha \cdot (-\beta) = -(\alpha\beta)$. Thus

$$\alpha\beta + \alpha\gamma = \alpha(\beta + \gamma).$$

The other cases are handled in the same way.

We have now completed the proof that $\mathbb{R}$ is an ordered field with the least upper bound property.

Why multiplication needs cases and addition does not

Addition is monotone in each argument on all of $\mathbb{Q}$: if r grows, so does $r + s$. Multiplication is not — multiplying by a negative reverses order. Since a cut is defined by *downward* closure, the union of products $\{rs\}$ points the wrong way as soon as a negative factor appears, and the Step 6 definition simply does not describe a cut.

Rudin's escape is to define multiplication only where it is well behaved, on $\mathbb{R}^+$, and then extend by the sign rules — which is the same device used at 1.16(c),(d) to prove $(-x)(-y) = xy$ from the field axioms. The cases in Step 7 are those rules read as definitions rather than theorems.

This asymmetry is why the appendix is longer than it looks. It is also why the distributive law, which mixes the two operations, must be checked case by case: β and γ can have opposite signs while $\beta + \gamma$ has either.

Step 8 We associate with each $r \in \mathbb{Q}$ the set r^* which consists of all $p \in \mathbb{Q}$ such that $p < r$. It is clear that each r^* is a cut; that is, $r^* \in \mathbb{R}$. These cuts satisfy the following relations:

- (a) $r^* + s^* = (r + s)^*$,
- (b) $r^* s^* = (rs)^*$,
- (c) $r^* < s^*$ if and only if $r < s$.

To prove (a), choose $p \in r^* + s^*$. Then $p = u + v$, where $u < r$, $v < s$. Hence $p < r + s$, which says that $p \in (r + s)^*$.

Conversely, suppose $p \in (r + s)^*$. Then $p < r + s$. Choose t so that $2t = r + s - p$, put

$$r' = r - t, \quad s' = s - t.$$

41 The trick is splitting the shortfall evenly. We know p falls short of $r + s$ by $2t > 0$; docking t from each of r and s keeps both strictly below their targets while making the sum land exactly on p . Halving the slack is the same device as $t = p + r/2$ in Step 4(A5).

Then $r' \in r^*$, $s' \in s^*$, and $p = r' + s'$, so that $p \in r^* + s^*$.⁴¹ This proves (a). The proof of (b) is similar.

If $r < s$ then $r \in s^*$, but $r \notin r^*$; hence $r^* < s^*$.

If $r^* < s^*$, then there is a $p \in s^*$ such that $p \notin r^*$. Hence $r \leq p < s$, so that $r < s$. This proves (c).

Step 9 We saw in Step 8 that the replacement of the rational numbers r by the corresponding “rational cuts” $r^* \in \mathbb{R}$ preserves sums, products, and order. This fact may be expressed by saying that the ordered field $\mathbb{Q}$ is *isomorphic* to the ordered field $\mathbb{Q}^*$ whose elements are the rational cuts. Of course, r^* is by no means the same as r , but the properties we are concerned with (arithmetic and order) are the same in the two fields.

It is this identification of $\mathbb{Q}$ with $\mathbb{Q}^*$ which allows us to regard $\mathbb{Q}$ as a subfield of $\mathbb{R}$.⁴²

The second part of Theorem 1.19 is to be understood in terms of this identification. Note that the same phenomenon occurs when the real numbers are regarded as a subfield of the complex field, and it also occurs at a much more elementary level, when the integers are identified with a certain subset of $\mathbb{Q}$.

It is a fact, which we will not prove here, that any two ordered fields with the least-upper-bound property are isomorphic. The first part of Theorem 1.19 therefore *characterizes* the real field $\mathbb{R}$ completely.⁴³

The books by Landau and Thurston cited in the Bibliography are entirely devoted to number systems. Chapter 1 of Knopp’s book contains a more leisurely description of how $\mathbb{R}$ can be obtained from $\mathbb{Q}$. Another construction, in which each real number is defined to be an equivalence class of Cauchy sequences of rational numbers (see Chap. 3), is carried out in Sec. 5 of the book by Hewitt and Stromberg.

The cuts in $\mathbb{Q}$ which we used here were invented by Dedekind. The construction of $\mathbb{R}$ from $\mathbb{Q}$ by means of Cauchy sequences is due to Cantor. Both Cantor and Dedekind published their constructions in 1872.⁴⁴

WHAT THE APPENDIX ESTABLISHED

$\mathbb{R}$ exists. Concretely, it is the set of downward-closed, unbounded-below, maximum-free subsets of $\mathbb{Q}$, ordered by inclusion, with addition given by elementwise sums and multiplication defined first for positives and then by sign rules. Under those definitions it is an ordered field, its least upper bounds are unions, and it contains an isomorphic copy of $\mathbb{Q}$.

Theorem 1.19 is therefore a theorem. And since complete ordered fields are unique up to isomorphism, this particular construction may now be forgotten — which is precisely what the rest of the book does.

[42] This settles the second sentence of Theorem 1.19, and it is worth being clear about what is and is not claimed. The rational 2 and the cut $2^* = \{p \in \mathbb{Q} : p < 2\}$ are genuinely different objects — one a number, the other an infinite set. What Step 8 establishes is that no arithmetic-or-order question can tell them apart, and we therefore agree to write $\mathbb{Q} \subset \mathbb{R}$. The same courtesy was extended to $\mathbb{R} \subset \mathbb{C}$ after 1.26.

[43] The strongest sentence in the appendix, and it is dropped without proof. It says that Dedekind’s construction was not one choice among many with different outcomes: any route to a complete ordered field — cuts, Cauchy sequences, infinite decimals — lands on the same object up to isomorphism. That is why nothing in the rest of the book ever refers back to cuts. $\mathbb{R}$ is fully determined by its properties, so the properties are all one needs.

[44] Cantor’s route is the one most modern texts take — Tao’s *Analysis I* builds $\mathbb{R}$ as equivalence classes of Cauchy sequences of rationals (his Chapter 5) and explicitly declines Dedekind cuts. It generalises: the same construction completes any metric space (Rudin does exactly this in Exercise 3.24), whereas cuts need an order and so stop at $\mathbb{R}$. Cuts are shorter here, which is why Rudin keeps them. Both books warn off the third route, infinite decimals: it looks familiar but has to legislate $0.999\cdots = 1$, and is the most delicate of the three.

EXERCISES

HOW TO USE THESE

Rudin's exercises are not drill. Several of them — 1.6 and 1.7 in particular — prove results the book will later use, and skipping them leaves a hole. The annotation here gives the *approach*: which theorem to reach for, what shape the argument takes, where the difficulty actually is. It does not give solutions. Read the note only after you are genuinely stuck, and never before attempting the problem.

The ones that matter most. 1.6 and 1.7 construct b^x for real x and its inverse, the logarithm — Rudin defers this to the exercises and then uses exponentials freely from Chapter 3 onward. 1.5 is the inf/sup duality you will use constantly. 1.8 explains why $\mathbb{C}$ is never given an order. 1.20 tests whether you understand why a cut may have no largest member.

The routine ones. 1.1–1.4 and 1.11–1.14 are short consequences of 1.14–1.18 and 1.33; do them quickly. 1.16–1.19 are geometry in $\mathbb{R}^k$ and exercise the Schwarz inequality.

45 Contradiction, and the point is that $\mathbb{Q}$ is closed under the operations. If $r+x$ were rational, then $x = (r+x) - r$ would be a difference of two rationals, hence rational. Same for rx , dividing by r — which is where $r \neq 0$ is needed.

46 The parity argument of 1.1 does not transfer directly, since 12 is not prime. Work with the factor 3 instead of 2: if $m^2 = 12n^2$ then $3 \mid m^2$, hence $3 \mid m$ (this needs 3 prime), and the descent proceeds as before.

47 The hypothesis “nonempty” is the entire exercise, and it is easy to write a proof that never uses it — such a proof is wrong. Pick any $x \in E$; then $\alpha \leq x \leq \beta$. If E were empty, every element would be both a lower and an upper bound and the conclusion would fail.

48 Reflection swaps the two notions: β is a lower bound of A if and only if $-\beta$ is an upper bound of $-A$, and the inequality reverses on negation. Verify both clauses of Definition 1.8 for $-\sup(-A)$. This is a second proof of 1.11 for the special case $S = \mathbb{R}$, and the identity is used without comment throughout Chapters 3 and 6. It is Tao's Exercise 5.5.1.

49 This is a construction, not a computation, and it is the exercise Rudin relies on most. The architecture: (a) shows the rational exponent is well defined — raise both sides to the power nq and use uniqueness in 1.21; (b) is then algebra; (c) is the crucial step, showing the new definition by supremum agrees with the old one on rationals, so that $b^x := \sup B(x)$ extends rather than contradicts; (d) is the payoff. Part (c) is where $b > 1$ matters — it makes $t \mapsto b^t$ increasing, so $B(r)$ has b^r as its largest element. If you get stuck, this construction is carried out in the main text of Tao's Analysis I, §5.6 and §6.7 — what Rudin relegates to an exercise, Tao develops as numbered propositions.

Unless the contrary is explicitly stated, all numbers that are mentioned in these exercises are understood to be real.

1.1. If r is rational ($r \neq 0$) and x is irrational, prove that $r+x$ and rx are irrational.⁴⁵

1.2. Prove that there is no rational number whose square is 12.⁴⁶

1.3. Prove Proposition 1.15.

1.4. Let E be a nonempty subset of an ordered set; suppose α is a lower bound of E and β is an upper bound of E . Prove that $\alpha \leq \beta$.⁴⁷

1.5. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that⁴⁸

$$\inf A = -\sup(-A).$$

1.6. Fix $b > 1$.⁴⁹

(a) If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

- (b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.
- (c) If x is real, define $B(x)$ to be the set of all numbers b^t , where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x .

- (d) Prove that $b^{x+y} = b^x b^y$ for all real x and y .

1.7. Fix $b > 1$, $y > 0$, and prove that there is a unique real x such that $b^x = y$, by completing the following outline. (This x is called the *logarithm* of y to the base b .)⁵⁰

- (a) For any positive integer n , $b^n - 1 \geq n(b - 1)$.
- (b) Hence $b - 1 \geq n(b^{1/n} - 1)$.
- (c) If $t > 1$ and $n > (b - 1)/(t - 1)$, then $b^{1/n} < t$.
- (d) If w is such that $b^w < y$, then $b^{w+(1/n)} < y$ for sufficiently large n ; to see this, apply part (c) with $t = y \cdot b^{-w}$.
- (e) If $b^w > y$, then $b^{w-(1/n)} > y$ for sufficiently large n .
- (f) Let A be the set of all w such that $b^w < y$, and show that $x = \sup A$ satisfies $b^x = y$.
- (g) Prove that this x is unique.

1.8. Prove that no order can be defined in the complex field that turns it into an ordered field. *Hint:* -1 is a square.⁵¹

1.9. Suppose $z = a + bi$, $w = c + di$. Define $z < w$ if $a < c$, and also if $a = c$ but $b < d$. Prove that this turns the set of complex numbers into an ordered set. (This type of order relation is called a *dictionary order*, or *lexicographic order*, for obvious reasons.) Does this ordered set have the least-upper-bound property?⁵²

1.10. Suppose $z = a + bi$, $w = u + iv$, and

$$a = \left(\frac{|w| + u}{2} \right)^{1/2}, \quad b = \left(\frac{|w| - u}{2} \right)^{1/2}.$$

Prove that $z^2 = w$ if $v \geq 0$ and that $(\bar{z})^2 = w$ if $v \leq 0$. Conclude that every complex number (with one exception!) has two complex square roots.⁵³

50 Compare the whole outline with the proof of 1.21 and the resemblance is exact: form the set of things that undershoot, take its supremum, and rule out both strict inequalities. Parts (a)–(e) are the continuity estimates that Rudin computed by hand there — (d) and (e) are the analogues of choosing h and k . Part (a) is the binomial theorem, or induction. Once you see that 1.21 and 1.7 are the same argument twice, the outline reads itself.

51 Two lines from 1.18(d). That proposition gives $x^2 > 0$ for $x \neq 0$ in any ordered field, so $i^2 = -1 > 0$; but $1 > 0$ also, and then (i) of 1.17 forces $0 = 1 + (-1) > 0$. The point is that no cleverness in choosing the order can help — the obstruction is 1.18(d), which holds for every order whatever.

52 Read alongside 1.8. An order on $\mathbb{C}$ certainly exists; what 1.8 rules out is an order compatible with the arithmetic. Definition 1.6 asks only for trichotomy and transitivity, and lexicographic order supplies both. For the second question, test the set $\{z : \operatorname{Re}(z) < 0\}$ and ask what its least upper bound would have to be.

53 Direct computation: expand $z^2 = a^2 - b^2 + 2abi$ and check the two parts separately. The real part is immediate; for the imaginary part you will need $|w|^2 = u^2 + v^2$ and a square root taken with the right sign, which is exactly where the hypothesis on v enters. Both a and b are well defined because $|w| \geq |u|$ by 1.33(d). The exception is $w = 0$.

54 Take $r = |z|$ and $w = z/|z|$, then handle $z = 0$ separately — and that separate case is the answer to the second question.

55 Induction on n , with 1.33(e) as the base and the inductive step. Worth doing once carefully: this is the form of the triangle inequality actually used in Chapter 3.

56 The reverse triangle inequality. Write $x = (x - y) + y$ and apply 1.33(e) to get $|x| - |y| \leq |x - y|$; then swap x and y to get the other direction, and combine. Used constantly later to show that the modulus is continuous.

57 Expand each term as $(1 \pm z)(\overline{1 \pm z})$ using 1.31, and watch the cross terms cancel. The answer is a constant — this is the parallelogram law of 1.17 in disguise.

58 Two spheres of radius r about $\mathbf{x}$ and $\mathbf{y}$: they meet in a sphere of dimension $k - 2$, a point, or nothing. Algebraically, start from the midpoint $\mathbf{m} = (\mathbf{x} + \mathbf{y})/2$ and look for $\mathbf{z} = \mathbf{m} + \mathbf{v}$ with $\mathbf{v}$ orthogonal to $\mathbf{x} - \mathbf{y}$; Exercise 1.17 gives $|\mathbf{v}|^2 = r^2 - d^2/4$. The condition $k \geq 3$ is what guarantees room for infinitely many such $\mathbf{v}$ — for $k = 2$ there are exactly two, for $k = 1$ none.

59 Expand both sides with the inner product; the cross terms $\pm 2\mathbf{x} \cdot \mathbf{y}$ cancel. Geometrically: the sum of the squares of a parallelogram's diagonals equals the sum of the squares of its four sides. This identity characterises norms that come from an inner product, which is why it fails for the norms of Exercise 1.19 in later chapters.

1.11. If z is a complex number, prove that there exists an $r \geq 0$ and a complex number w with $|w| = 1$ such that $z = rw$. Are w and r always uniquely determined by z ?⁵⁴

1.12. If $z_1, \dots, z_n$ are complex, prove that⁵⁵

$$|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|.$$

1.13. If x, y are complex, prove that⁵⁶

$$||x| - |y|| \leq |x - y|.$$

1.14. If z is a complex number such that $|z| = 1$, that is, such that $z\bar{z} = 1$, compute⁵⁷

$$|1 + z|^2 + |1 - z|^2.$$

1.15. Under what conditions does equality hold in the Schwarz inequality?

1.16. Suppose $k \geq 3$, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, $|\mathbf{x} - \mathbf{y}| = d > 0$, and $r > 0$. Prove:

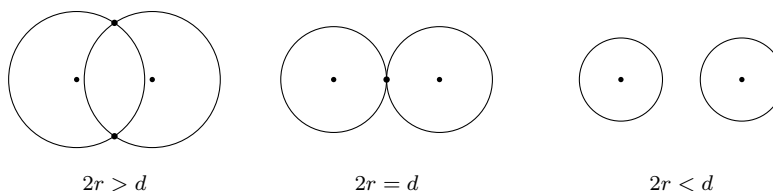
(a) If $2r > d$, there are infinitely many $\mathbf{z} \in \mathbb{R}^k$ such that

$$|\mathbf{z} - \mathbf{x}| = |\mathbf{z} - \mathbf{y}| = r.$$

(b) If $2r = d$, there is exactly one such $\mathbf{z}$.

(c) If $2r < d$, there is no such $\mathbf{z}$.

How must these statements be modified if k is 2 or 1?⁵⁸

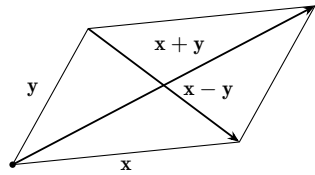


Drawn in the plane, where the intersection is two points, one, or none. In $\mathbb{R}^k$ the first case becomes a sphere of dimension $k - 2$ — a circle when $k = 3$, and infinite whenever $k \geq 3$.

1.17. Prove that

$$|\mathbf{x} + \mathbf{y}|^2 + |\mathbf{x} - \mathbf{y}|^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2$$

if $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^k$. Interpret this geometrically, as a statement about parallelograms.⁵⁹



sum of squares
of the diagonals
= sum of squares
of the four sides

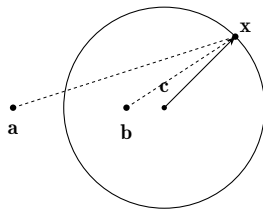
The identity is exactly this, and it fails for norms that do not come from an inner product.

1.18. If $k \geq 2$ and $\mathbf{x} \in \mathbb{R}^k$, prove that there exists $\mathbf{y} \in \mathbb{R}^k$ such that $\mathbf{y} \neq \mathbf{0}$ but $\mathbf{x} \cdot \mathbf{y} = 0$. Is this also true if $k = 1$?⁶⁰

1.19. Suppose $\mathbf{a} \in \mathbb{R}^k$, $\mathbf{b} \in \mathbb{R}^k$. Find $\mathbf{c} \in \mathbb{R}^k$ and $r > 0$ such that

$$|\mathbf{x} - \mathbf{a}| = 2|\mathbf{x} - \mathbf{b}|$$

if and only if $|\mathbf{x} - \mathbf{c}| = r$. (Solution: $3\mathbf{c} = 4\mathbf{b} - \mathbf{a}$, $3r = 2|\mathbf{b} - \mathbf{a}|$.)⁶¹



every $\mathbf{x}$ with
 $|\mathbf{x} - \mathbf{a}| = 2|\mathbf{x} - \mathbf{b}|$
lies on one sphere

The locus of points whose distances to two fixed points keep a constant ratio is a sphere — an Apollonius sphere — not, as one might guess, a plane. It is a plane only when the ratio is 1.

1.20. With reference to the Appendix, suppose that property (III) were omitted from the definition of a cut. Keep the same definitions of order and addition. Show that the resulting ordered set has the least-upper-bound property, that addition satisfies axioms (A1) to (A4) (with a slightly different zero-element!), but that (A5) fails.

On Exercise 1.20

The exercise that explains a definition rather than applying it. Without (III) each rational r has *two* representatives, $\{p < r\}$ and $\{p \leq r\}$; the parenthetical hint is that the identity is now $\{p \leq 0\}$, not $\{p < 0\}$. Inverses are what break. Take $\alpha = \{p < 0\}$, which has no largest member: then $\alpha + \beta$ has no largest member either, whatever β is, so it can never equal the new zero — which has largest member 0. Compare the note at Step 1.

60 Exhibit one: if $\mathbf{x} = (x_1, \dots, x_k)$, try $\mathbf{y} = (-x_2, x_1, 0, \dots, 0)$, and check separately what to do when $x_1 = x_2 = 0$. For $k = 1$ the statement fails in general but not always: if $x \neq 0$ then $xy = 0$ forces $y = 0$ by 1.16(b), so no such y exists — whereas if $x = 0$ then every $y \neq 0$ works. So the honest answer is that it holds precisely when $x = 0$.

61 Square both sides, expand with the inner product, and collect: the $|\mathbf{x}|^2$ terms do not cancel — they combine to $-3|\mathbf{x}|^2$ — so dividing by -3 leaves the equation of a sphere. Completing the square identifies the centre. The locus is an Apollonius sphere; Rudin supplies the answer so you can check your algebra.

Appendix to Chapter 1

HOW TO USE THIS APPENDIX

Nothing in this appendix is Rudin's text. It is imported from the four comparison texts and set apart at every level: the running head names the appendix, section heads carry a reverse-video tag, item numbers are prefixed **S**, and every statement names its source.

Why import anything at all. Chapter 1 is the one chapter where Rudin is deliberately incomplete, and he says so. He assumes $\mathbb{Q}$ outright, exiles the construction of $\mathbb{R}$ to an appendix he tells you to skip, and states his sharpest result — that the complete ordered field is unique — with the words “*which we will not prove here.*” Each omission is defensible; each also leaves a question a reader is entitled to have answered.

What is here.

A. The uniqueness theorem, proved. Rudin's strongest sentence about $\mathbb{R}$ is the one he declines to justify. Proving it is what converts his appendix from “one construction” into “the only possible answer.”

B. How much is the Archimedean property? Rudin gets it free from LUB at 1.20(a), so he never has to show it is not automatic. It is not: there are ordered fields containing infinite elements, and seeing one explains why Yu Pin must assume Archimedes separately.

C. Decimals, and a third route to the supremum. Rudin disposes of decimals in three lines and says he will never use them. ECNU *defines* reals as decimals, which makes $0.999\dots = 1$ a normalisation rule and turns the supremum into something one computes rather than asserts.

D. The toolkit Rudin never states. The real triangle inequality, its reverse form, the band rule for absolute values, and the ε -principle — four facts he uses constantly and states nowhere.

E. Below Rudin's starting line. He assumes $\mathbb{Q}$ in his second sentence. Tao pays that bill in full, and the itemised cost explains what Rudin's appendix is an instance of.

F. $\mathbb{R}$ is not the only completion of $\mathbb{Q}$. “Filling the holes” presumes a way of measuring closeness. Change the absolute value and the holes move.

G. Real exponents, worked. Rudin’s Exercises 1.6–1.7 done properly, including the two steps his hint glosses.

How to read it. Section A belongs with Rudin’s own appendix, and is best read straight after it. Section B can be read as soon as 1.20 is done, C alongside 1.22, D before Chapter 3, and G with Exercises 1.6–1.7. E and F are optional and answer “what did the assumptions cost?” The exercises are the last section.

NOT IN RUDIN

SUPPLEMENT A — THE UNIQUENESS THEOREM, PROVED

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.19 asserts that an ordered field $\mathbb{R}$ with the least-upper-bound property exists and contains $\mathbb{Q}$ as a subfield; his appendix builds one from Dedekind cuts. At the end of that appendix he writes: “*It is a fact, which we will not prove here, that any two ordered fields with the least-upper-bound property are isomorphic*” — and concludes that 1.19 characterizes $\mathbb{R}$ completely.

The gap. That sentence is the load-bearing one for the entire book, and it arrives unsupported. Everything after Chapter 1 uses only the *properties* of $\mathbb{R}$ and never the cuts; the licence for that is precisely uniqueness. Without it, a sceptic could ask whether the theorems of Chapters 2–11 are theorems about *Dedekind’s* real numbers rather than about the real numbers.

Where the gap bites. Immediately and permanently. It is why Rudin can tell the reader, in his Preface, to skip the appendix; why Tao may build $\mathbb{R}$ from Cauchy sequences and get the same object; why ECNU may define reals as infinite decimals; and why the six equivalent completeness axioms of the Chapter 1 box are equivalent *as descriptions of one thing* rather than as six different subjects.

What this section supplies. The proof. It is not hard — it is the density of $\mathbb{Q}$ used four times — but it is the argument that retires the construction for good.

S1.1 Theorem. *Let F_1 and F_2 be ordered fields with the least-upper-bound property. Then there is exactly one order-preserving field isomor-*

phism $\varphi : F_1 \rightarrow F_2$, and it fixes the rationals.

SOURCE: Nobody's. Rudin states it and declines to prove it; Tao never introduces the notion of an ordered field at all; Yu Pin defers it (p. 35, "as for uniqueness, we will discuss it in passing later") and delivers only the metric version — the completion of a metric space is unique up to isometry (p. 124). The proof below is supplied here.

Proof. Each of F_1, F_2 contains an isomorphic copy of $\mathbb{Q}$, obtained by sending $n \mapsto 1 + \cdots + 1$ and then $m/n \mapsto m \cdot n^{-1}$; the field axioms make this injective and order-preserving, and we identify $\mathbb{Q}$ with its image in each. By 1.20, applied in each field, both are archimedean and $\mathbb{Q}$ is dense in each.

For $x \in F_1$ put

$$Q_x = \{ q \in \mathbb{Q} : q < x \}, \quad \varphi(x) = \sup_{F_2} Q_x.$$

The set Q_x is nonempty and bounded above in F_2 : by the archimedean property there are integers $m < x < n$, so $m \in Q_x$ and n bounds it. Hence the supremum exists in F_2 by hypothesis, and φ is defined.

φ fixes $\mathbb{Q}$ and preserves order. If $r \in \mathbb{Q}$ then $Q_r = \{ q \in \mathbb{Q} : q < r \}$ has supremum r in F_2 , so $\varphi(r) = r$. If $x < y$ in F_1 , density gives rationals with $x < q < r < y$; then every element of Q_x is below q while $r \in Q_y$, so $\varphi(x) \leq q < r \leq \varphi(y)$. In particular φ is injective.

φ is additive. $Q_x + Q_y \subset Q_{x+y}$ gives $\varphi(x) + \varphi(y) \leq \varphi(x+y)$. Conversely, if $q < x+y$ is rational, density supplies rationals $q_1 < x$ and $q_2 < y$ with $q < q_1 + q_2$ — choose q_1 within $\frac{1}{2}(x+y-q)$ of x and q_2 likewise — so $q \leq \varphi(x) + \varphi(y)$. Taking the supremum over such q gives the reverse inequality.

φ is multiplicative. For $x, y > 0$ the same two-sided argument applies to products of positive rationals; the general case follows from additivity, since $\varphi(-x) = -\varphi(x)$.

φ is onto. Given $z \in F_2$, set $x = \sup_{F_1} \{ q \in \mathbb{Q} : q < z \}$, which exists for the same reason as before. Then Q_x and $\{ q \in \mathbb{Q} : q < z \}$ have the same elements, by density in each field, so $\varphi(x) = z$.

Uniqueness of φ . Any order-preserving field isomorphism fixes 1, hence fixes $\mathbb{Q}$; and an order-preserving map fixing a dense set is determined, since $\psi(x)$ must be an upper bound of $\psi(Q_x) = Q_x$ and below every rational above x . $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** An isomorphism must be *built*, and the only structure both fields are known to share is $\mathbb{Q}$ and the supremum operation. So the map must be defined by naming each x with rationals and reassembling it on the other side: pattern 3 of Chapter 1's strategy index, transport to the supremum.
2. **Buy the supremum.** $\varphi(x) = \sup Q_x$ needs Q_x nonempty and bounded — exactly the two tickets 1.10 demands, and exactly what the archimedean property (1.20(a)) supplies. Note this is

the first use of LUB, and it is used to *define* the map, not to prove anything about it.

3. **Get order for free.** Density (1.20(b)) separates any two points by two rationals, and two rationals are enough to separate their images. Injectivity comes with it.
4. **Do the algebra by squeezing.** Additivity and multiplicativity are each proved by two inequalities: one inclusion of sets gives “ $\leq$,” and density gives “ $\geq$.” There is no computation anywhere — only the fact that rationals approximate from below as closely as one likes.
5. **Run the construction backwards for surjectivity.** The same formula with the roles of F_1 and F_2 exchanged produces the preimage. Symmetry of the hypotheses is what makes this free.
6. **Audit.** The least-upper-bound property is spent twice, once in each field, both times to define a map. The archimedean property and density of $\mathbb{Q}$ are spent repeatedly — but both are *consequences* of LUB by 1.20, so the hypothesis list is exactly Rudin’s. Nothing about cuts, decimals or Cauchy sequences appears anywhere, which is the whole point.

What the theorem licenses

Read the theorem as a permission slip. It says the phrase “the real numbers” denotes a single object, so that:

Constructions are disposable. Rudin’s cuts, Tao’s Cauchy sequences, and ECNU’s infinite decimals are three routes to the same destination. Rudin’s advice in the Preface — that beginning with a construction is “pedagogically unsound (though logically correct)” — is only defensible because of this theorem: you may skip the construction precisely because the construction does not matter.

Axiom choices are disposable too. Chapter 1’s box compared four axiomatisations — Rudin’s LUB, Yu Pin’s nested intervals, Tao’s Cauchy completeness, ECNU’s six-cycle. Uniqueness is what makes those four descriptions of one field rather than four fields.

And it is sharp. Drop any hypothesis and the conclusion fails. Without LUB there are many non-isomorphic archimedean ordered fields — $\mathbb{Q}$ itself, and every subfield of $\mathbb{R}$. Without the ordering there are field isomorphisms of $\mathbb{C}$ that are wildly discontinuous. The next section drops the archimedean property, which LUB was silently supplying, and shows what escapes.

SUPPLEMENT B — HOW MUCH IS THE ARCHIMEDEAN PROPERTY?

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.20(a) derives the archimedean property — for $x > 0$ and any y there is a positive integer n with $nx > y$ — from the least-upper-bound property, in four lines. His margin remark notes that this is a theorem, not an axiom.

The gap. Deriving a property is not the same as showing it has content. Because Rudin obtains Archimedes free of charge, a reader has no way to see that it is a genuine restriction — that ordered fields exist in which some element exceeds every integer, and that such fields therefore cannot have the LUB property.

Where the gap bites. At Chapter 1's own comparison box, where Yu Pin axiomatises $\mathbb{R}$ as field + order + *archimedean* + nested intervals, and the natural question — why does he need the third hypothesis when Rudin does not? — has no answer inside Rudin. It bites again at S2.7 in the Chapter 2 appendix, where recovering sup from Cauchy completeness needs Archimedes as a separate hypothesis, for exactly the reason exhibited here.

What this section supplies. An explicit non-archimedean ordered field, small enough to hold in the hand, together with the proof that no such field can have the least-upper-bound property.

S1.2 Example. (An ordered field with infinite elements.) Let $\mathbb{R}(t)$ be the field of rational functions $p(t)/q(t)$ with real coefficients, and declare $f > 0$ when $f(x) > 0$ for all sufficiently large real x — equivalently, when the leading coefficients of p and q have the same sign. This makes $\mathbb{R}(t)$ an ordered field in which

$$t > n \quad \text{for every positive integer } n.$$

SOURCE: Supplied here: an explicit non-archimedean ordered field is absent from all four comparison texts. Yu Pin's axiom list (field, order, archimedean, nested intervals) is built to expose the possibility — he notes at p. 26 that (F), (O) and (A) alone are satisfied by $\mathbb{Q}$ — but he gives no example in which (A) fails.

CHECKING THE ORDER, AND READING THE ELEMENT T

That the relation is an order needs only that a nonzero rational function has constant sign near $+\infty$ — it has finitely many zeros — and that the sum and product of two eventually-positive functions are eventually positive. Both are immediate, and they are the two clauses of Definition 1.17.

Now look at t . For any integer n the function $t - n$ is eventually positive, so $t > n$: the single element t exceeds every integer. Its

reciprocal $1/t$ is then a positive element smaller than every $1/n$ — a genuine infinitesimal. Neither is a limit of anything; they are simply elements of the field, and the field’s arithmetic is entirely ordinary.

The moral is that “ $\mathbb{R}$ has no infinite or infinitesimal elements” is not a consequence of being an ordered field. It has to be imposed, and Rudin imposes it without naming it, by assuming LUB.

S1.3 Theorem. *An ordered field with the least-upper-bound property is archimedean. Consequently $\mathbb{R}(t)$ does not have the least-upper-bound property: the set of positive integers is bounded above in $\mathbb{R}(t)$ but has no supremum.*

Proof. This is Rudin’s 1.20(a) restated, and his proof is the argument. Suppose the set $\mathbb{N}$ of positive integers were bounded above in an ordered field with the LUB property, and let $\alpha = \sup \mathbb{N}$. Since $\alpha - 1 < \alpha$, the element $\alpha - 1$ is not an upper bound, so $n > \alpha - 1$ for some $n \in \mathbb{N}$; but then $n + 1 > \alpha$ and $n + 1 \in \mathbb{N}$, contradicting that α is an upper bound.

Hence in such a field $\mathbb{N}$ is unbounded, which is the archimedean property. In $\mathbb{R}(t)$ the element t bounds $\mathbb{N}$ above, so $\mathbb{R}(t)$ cannot have the LUB property. $\square$

Three fields, and what separates them

Line up the examples Chapter 1 now has:

field	arch.	LUB	what it lacks
$\mathbb{Q}$	yes	no	the sup of $\{p : p^2 < 2\}$ (1.1)
$\mathbb{R}(t)$	no	no	any bound on $\mathbb{N}$
$\mathbb{R}$	yes	yes	—

$\mathbb{Q}$ and $\mathbb{R}(t)$ fail to be $\mathbb{R}$ for genuinely different reasons, and the difference is worth holding on to. $\mathbb{Q}$ has the right *scale* — it is archimedean, its integers are unbounded — and is merely full of holes. $\mathbb{R}(t)$ has no holes of $\mathbb{Q}$ ’s kind at all in the relevant sense; it fails earlier and more crudely, by containing elements no integer can reach.

This is why Yu Pin’s axiom list needs four items where Rudin’s needs one extra property. Nested intervals alone does not imply Archimedes — one can have a non-archimedean field in which every nested sequence of intervals has a common point — so he must assume it. Rudin’s LUB implies both at once, which is economical but conceals that two separate things were bought.

NOT IN RUDIN

SUPPLEMENT C — DECIMALS, AND A THIRD ROUTE TO THE SUPREMUM

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.22 disposes of decimals in three lines: given $x > 0$, take the largest integer $n_0 \leq x$, then the largest n_1 with $n_0 + n_1/10 \leq x$, and so on; the resulting decimal converges to x . He then writes that “*we shall never use decimals*” and does not mention them again.

The gap. Two, in opposite directions. Forwards: his appendix builds $\mathbb{R}$ from cuts and stops, never reconnecting the constructed field to the notation every reader has used since childhood — so “what *is* $0.999\dots$?” is never answered. Backwards: he presents LUB as an axiom and its construction as a one-off, leaving the impression that a supremum is an object one may assert but not compute.

Where the gap bites. At 1.22 itself, where the reader is told the decimal expansion exists and is unique but not what makes the uniqueness work; at the end of the appendix, where cuts are abandoned without ever being converted into anything familiar; and in every later argument that approximates a real by rationals — 1.20(b), 1.21, 3.20 — each of which rebuilds an approximation scheme that 1.22 could have supplied once.

What this section supplies. ECNU’s development, which runs the other way: reals *are* infinite decimals, the awkward identification $0.999\dots = 1$ is a normalisation rule chosen in advance, and the supremum is produced by an algorithm — one digit at a time — rather than asserted.

S1.4 Definition. (*Reals as decimals.*) A real number is an infinite decimal $a_0.a_1a_2a_3\dots$; it is rational when the expansion terminates or eventually repeats and irrational otherwise. To make the representation unique, every terminating expansion is rewritten with a tail of nines: for $a_n \neq 0$,

$$a_0.a_1\dots a_n = a_0.a_1\dots(a_n - 1)999\dots,$$

and an integer a_0 is written $(a_0 - 1).999\dots$. Order is then lexicographic: $x > y$ if $a_0 > b_0$, or if the digits agree up to place l and $a_{l+1} > b_{l+1}$.

SOURCE: ECNU 定义 1 (bk. pp. 1–2), where this is the definition of $\mathbb{R}$, not a theorem about it.

$0.999\dots = 1$ IS A CONVENTION, NOT A PARADOX

Students meet this identity as a puzzle. In ECNU’s development it is not a puzzle at all: $0.999\dots$ is the *official name* of 1, chosen so that the lexicographic order of Definition S1.4 is a genuine total order.

Why the choice matters: without a convention, 2.001 and $2.000999\dots$ are two names for one number, and lexicographic comparison of names would declare the first larger. Normalising every terminating decimal

to its nine-tail leaves exactly one name per number, and comparison of names then computes comparison of numbers. Note the convention runs *opposite* to intuition — the canonical form is the infinite one.

Rudin sidesteps all of this by refusing to use decimals. His 1.22 asserts existence and uniqueness of the expansion and moves on; the uniqueness there is bought by taking the *largest* n_k at each stage, which produces the terminating form and so is the opposite normalisation. Neither choice is better — but a reader who has seen only one will find the other's $0.999\dots$ baffling.

S1.5 Definition. (*Deficient and excess approximations.*) For $x = a_0.a_1a_2\dots \geq 0$ put

$$x_n = a_0.a_1\dots a_n, \quad \bar{x}_n = x_n + 10^{-n},$$

so that $x_n \leq x < \bar{x}_n$ with $\bar{x}_n - x_n = 10^{-n}$. The x_n increase and the $\bar{x}_n$ decrease.

SOURCE: ECNU 定义 2 (bk. p. 2).

S1.6 Theorem. $x > y$ if and only if $x_n > \bar{y}_n$ for some n . Consequently, if $x < y$ there is a rational r with $x < r < y$.

SOURCE: ECNU 命题 and 例 1 (bk. pp. 2–3).

Proof. If $x_n > \bar{y}_n$ then $x \geq x_n > \bar{y}_n > y$. Conversely, if $x > y$ then $x - y > 0$, and since $\bar{x}_n - x_n = 10^{-n} \rightarrow 0$ the sandwich $x_n \leq x$, $y < \bar{y}_n$ forces $x_n > \bar{y}_n$ once 10^{-n} is smaller than the gap.

For density, apply this with the roles reversed: $x < y$ gives an n with $\bar{x}_n < y_n$, and

$$r = \frac{1}{2}(\bar{x}_n + y_n)$$

is rational — being the average of two terminating decimals — with $x \leq \bar{x}_n < r < y_n \leq y$. $\square$

Two proofs of density, and what each costs

Compare with Rudin's 1.20(b). He proves density from the archimedean property: stretch the interval by an integer n until $n(y - x) > 1$, catch an integer m in the stretched gap, and divide back. The catching step needs that a set of integers bounded below has a least element — well-ordering — which he uses without comment.

The proof above needs none of that. It truncates both numbers at a common depth and averages, and the only fact used is that the truncations sandwich the number within 10^{-n} . It is also *constructive*: it names the rational, whereas Rudin's argument asserts that a suitable m exists.

The trade is that this proof is available only once reals are decimals. Rudin's works in any archimedean ordered field, which is what he needs, since at 1.20 he has no decimals and does not want any.

S1.7 Theorem. (*The supremum, computed.*) Every nonempty set $S \subset \mathbb{R}$ that is bounded above has a least upper bound, and its decimal digits can be generated one at a time.

SOURCE: ECNU 确界原理 定理 1.1 (bk. pp. 5–7), proved from the decimal definition.

Proof. Assume S contains a nonnegative number. Choose a nonnegative integer n such that $x < n + 1$ for every $x \in S$ while some $a_0 \in S$ has $a_0 \geq n$; this is possible because S is bounded above and nonempty.

Divide $[n, n + 1)$ into ten equal parts. Among the ten, there is a largest digit n_1 for which some element of S reaches $n.n_1$, and then

$$x < n.n_1 + 10^{-1} \quad \text{for all } x \in S, \quad a_1 \geq n.n_1 \quad \text{for some } a_1 \in S.$$

Divide $[n.n_1, n.n_1 + 10^{-1})$ into ten parts and repeat. Inductively one obtains digits $n_1, n_2, \dots$ with

$$x < n.n_1 \cdots n_k + 10^{-k} \quad (x \in S), \quad a_k \geq n.n_1 \cdots n_k \quad \text{for some } a_k \in S. \quad (\text{S1.1})$$

Let $\eta = n.n_1 n_2 n_3 \cdots$ be the real number those digits name.

η is an upper bound: if some $x \in S$ had $x > \eta$, then by S1.6 $x_k > \bar{\eta}_k = n.n_1 \cdots n_k + 10^{-k}$ for some k , contradicting the first half of (S1.1).

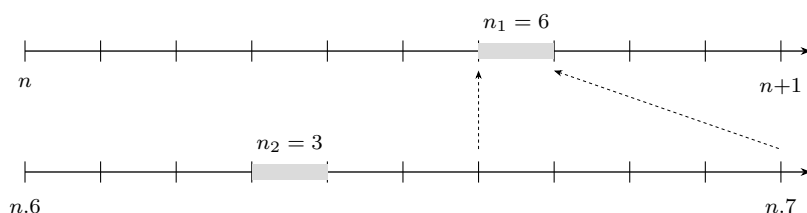
η is least: if $\alpha < \eta$, then by S1.6 there is k with $\eta_k > \bar{\alpha}_k$, and the second half of (S1.1) supplies $a_k \in S$ with $a_k \geq \eta_k > \bar{\alpha}_k \geq \alpha$, so α is not an upper bound. $\square$

HOW THE PROOF FLOWS

1. **Choose the method.** Unlike Rudin, we may not assume the supremum exists — it is the conclusion. So it must be *constructed*, and the only construction material available is decimal digits. The proof is therefore an algorithm: decide the digits one at a time.
2. **Fix the integer part.** Boundedness above gives an integer ceiling; nonemptiness gives a floor. Both hypotheses of 1.10 are spent here, in the first sentence, exactly as in Rudin’s 1.21.
3. **Choose each digit greedily.** At each stage take the largest digit that some element of S still reaches. “Largest that is still attained” is what makes the result least; “still an upper bound” is what makes it an upper bound. The two clauses of (S1.1) are the two halves of Definition 1.8, maintained as loop invariants.
4. **Let the algorithm run forever.** The digits define a real number by S1.4 — and this is the step that would fail over $\mathbb{Q}$, since the digit string need not repeat.
5. **Verify both clauses.** Each is checked by S1.6, which converts a comparison of reals into a comparison of finite truncations.

The invariants were designed so that this conversion closes the argument immediately.

- 6. Audit.** Boundedness and nonemptiness are spent once each, in step 2. The decimal definition of $\mathbb{R}$ is spent once, in step 4. Nothing else enters — in particular no cuts, no Cauchy sequences, and no appeal to LUB, which is what makes this a genuine *third* construction rather than a restatement.



each stage magnifies the chosen tenth and picks the next digit

The supremum computed digit by digit: at each stage the interval is cut into ten, and the algorithm takes the rightmost part that some element of S still reaches. The two conditions (S1.1) — still an upper bound, still attained — are the two clauses of Definition 1.8 carried along as invariants, and the digits they force out spell $\sup S$.

NOT IN RUDIN

SUPPLEMENT D — THE TOOLKIT RUDIN NEVER STATES

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. He proves the triangle inequality twice — at 1.33(e) for complex numbers and at 1.37(e) for $\mathbb{R}^k$ — and defines neighbourhoods once, at 2.18(a), as open balls.

The gap. The real-line versions of both are missing. The real triangle inequality never appears as a standalone statement; the reverse form $||a| - |b|| \leq |a - b|$ appears nowhere at all; the rule that converts $|a| < h$ into a two-sided band is never written down; and the ε -principle — if $a < b + \varepsilon$ for every $\varepsilon > 0$ then $a \leq b$ — is used in almost every proof from Chapter 1 onward without ever being stated.

Where the gap bites. Every ε -argument in the book opens by unpacking an absolute value into a band and closes by letting ε shrink. Rudin performs both moves silently — at 1.20, 1.21, 3.2, 3.3, 4.2 — so a reader learns them by imitation rather than by statement, and the reverse triangle inequality, which is what one needs to bound a difference of norms from below, has to be rediscovered each time it is wanted.

What this section supplies. The four facts, stated once, with

the proof of the triangle inequality that derives it from the band rule rather than by cases.

S1.8 Theorem. (*Absolute value.*) For real a, b and $h > 0$:

- (a) $-|a| \leq a \leq |a|$;
- (b) $|a| < h$ if and only if $-h < a < h$;
- (c) $||a| - |b|| \leq |a \pm b| \leq |a| + |b|$.

SOURCE: ECNU §1.2, six properties of $|\cdot|$ (bk. p. 3).

Proof. (a) and (b) are the definition read in the two cases $a \geq 0$ and $a < 0$.

For (c), add the two instances of (a):

$$-(|a| + |b|) \leq a + b \leq |a| + |b|,$$

and apply (b) in the reverse direction to read that band as $|a + b| \leq |a| + |b|$. Replacing b by $-b$ gives the same bound for $|a - b|$. For the left-hand inequality, write $|a| = |(a - b) + b| \leq |a - b| + |b|$, so $|a| - |b| \leq |a - b|$; exchanging a and b gives $|b| - |a| \leq |a - b|$, and the two together are the claim. $\square$

WHY (B) IS THE LOAD-BEARING ONE

Parts (a) and (c) are the famous ones, but (b) is the one used most often and the one Rudin never states. It is the rule that turns a statement about a *distance* into a statement about *two inequalities*, and back again — and the proof above shows the “back again” direction doing real work: the triangle inequality is proved by producing a band and then reading it as an absolute value, with no case analysis anywhere.

Every ε - δ argument in the book is an application. When Rudin writes “ $|s_n - s| < \varepsilon$ ” and then concludes “ $s - \varepsilon < s_n < s + \varepsilon$,” that is (b), used without comment.

S1.9 Theorem. (*The ε -principle.*) If $a < b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$. Equivalently, if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$.

SOURCE: ECNU 例 2 (bk. p. 3) and 习题 1.1 #3.

Proof. Suppose $a > b$. Then $\varepsilon = a - b$ is positive, and the hypothesis applied to this particular ε gives $a < b + (a - b) = a$, which is absurd. $\square$

The proof is three lines and the statement is used everywhere, which is exactly why it deserves a name.⁶²

S1.10 Definition. (*Neighbourhood system.*) For $a \in \mathbb{R}$ and $\delta > 0$ write

$$U(a; \delta) = \{x : |x - a| < \delta\}, \quad U^\circ(a; \delta) = \{x : 0 < |x - a| < \delta\},$$

62 Watch for it in Rudin. The last step of 1.11’s proof, of 1.21, of 3.2(b), and of every uniqueness argument in Chapter 3 is this principle: having shown a quantity is smaller than every positive number, conclude it is zero. Rudin never separates the move from its instances, so it reads as a trick repeated rather than a theorem applied.

and $U_+(a; \delta) = [a, a + \delta)$, $U_-(a; \delta) = (a - \delta, a]$ for the one-sided versions; $U(+\infty) = \{x : x > M\}$, $U(-\infty) = \{x : x < -M\}$ and $U(\infty) = \{x : |x| > M\}$ for M large.

SOURCE: ECNU §2.1 (bk. pp. 4–5).

One notation, and the four definitions it collapses

Rudin introduces neighbourhoods once, at 2.18(a), and only the two-sided ball. Every limit notion he later defines is then written out separately: 4.1 for $\lim_{x \rightarrow p}$, 4.25 for one-sided limits, 4.33 for limits at $\pm\infty$. The three definitions look unrelated on the page.

With the system above they are one sentence. In every case, $\lim f = A$ means

for each $U(A; \varepsilon)$ there is $U^\circ(a; \delta)$ with $f(U^\circ(a; \delta)) \subset U(A; \varepsilon)$,

and the four notions differ only in which neighbourhood of a is used: punctured, right-hand, left-hand, or a neighbourhood of $+\infty$. The Chapter 4 appendix reaches the same unification from the other direction, through Tao’s “limit along a subset” (S4.2); ECNU gets there by naming the neighbourhoods instead of the index sets.

NOT IN RUDIN

SUPPLEMENT E — BELOW RUDIN’S STARTING LINE

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. The second sentence of the book fixes his starting line: “*we shall not, however, enter into any discussion of the axioms that govern the arithmetic of the integers, but assume familiarity with the rational numbers.*” Everything in Chapter 1 is built on $\mathbb{Q}$, taken as given.

The gap. A reader cannot tell whether that assumption is cheap or expensive. Is $\mathbb{Q}$ a few lines of set theory, or a serious construction? And if it is serious, is Rudin’s Dedekind-cut appendix a special trick or an instance of something standard?

Where the gap bites. At the appendix, which looks like a one-off contrivance — why *sets of rationals*, of all things? — and at every use of induction in the book, since Rudin never says what property of $\mathbb{N}$ makes induction legitimate.

What this section supplies. Tao’s construction of $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$, priced honestly; and the observation that his three constructions and Rudin’s cuts are the same manoeuvre performed four times.

S1.11 Definition. (*The Peano axioms.*) $\mathbb{N}$ is a set with a distinguished element 0 and a successor operation $n \mapsto n++$ satisfying: 0 is not a

successor; the successor operation is injective; and if $P(0)$ holds and $P(n)$ implies $P(n++)$ for every n , then $P(n)$ holds for all n .

SOURCE: Tao I, Axioms 2.1–2.5 (bk. pp. 16–21).

What Rudin’s one sentence costs

Tao pays the bill in full, and the itemisation is instructive.

Even addition is a theorem first. Before $+$ can be defined by recursion one must prove that recursive definitions are legitimate at all — Tao’s Proposition 2.1.16. Induction is what licenses it. This is the answer to “what makes $\mathbb{N}$ special”: not that its elements are small, but that it supports recursion.

$\mathbb{Z}$ is a quotient. An integer is a formal difference $a - b$ of naturals, with $a - b = c - d$ declared to mean $a + d = c + b$. One then checks that $+$, $\times$ and $<$ are well defined on the classes.

$\mathbb{Q}$ is another quotient. A rational is a formal quotient $a//b$ with $b \neq 0$, and $a//b = c//d$ means $ad = cb$. Same checking again.

So Rudin’s “assume familiarity with the rational numbers” is roughly thirty-five pages, two quotient constructions and a theorem about recursion. His Preface defends skipping it as avoiding what is “pedagogically unsound (though logically correct),” and after S1.1 — uniqueness — that defence is airtight: the construction cannot matter, so it may be skipped.

ONE MANOEUVRE, PERFORMED FOUR TIMES

Look at what $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ have in common as constructions:

to build	formal symbol	identified when
$\mathbb{Z}$ from $\mathbb{N}$	$a - b$	$a + d = c + b$
$\mathbb{Q}$ from $\mathbb{Z}$	$a//b$	$ad = cb$
$\mathbb{R}$ from $\mathbb{Q}$ (Tao)	$\text{LIM } a_n$	$\lim a_n - b_n = 0$
$\mathbb{R}$ from $\mathbb{Q}$ (Rudin)	a cut α	— (the set <i>is</i> the number)

The first three are identical in shape: introduce a symbol for the operation you wish you could perform, declare when two symbols name the same thing, and prove the arithmetic descends to the classes. Tao says so explicitly — LIM is introduced as “a new formal symbol, similar to $-$ and $//$ ” — and adds that once formal limits are shown to be actual limits, they are never needed again. Rudin’s cuts are the odd one out: a cut is not a symbol standing for something absent but a set that *is* the number. That is why his appendix needs no equivalence classes and why Step 1 spends so long on property (III) — the normalisation that makes the representation unique is built into the definition rather than quotiented out afterwards.

SUPPLEMENT F — $\mathbb{R}$ IS NOT THE ONLY COMPLETION OF $\mathbb{Q}$

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Section 1.1 exhibits the gap in $\mathbb{Q}$ — the set $\{p : p^2 < 2\}$ with no rational supremum — and 1.19 supplies $\mathbb{R}$ as the field that fills it. The narrative is: $\mathbb{Q}$ has holes, $\mathbb{R}$ is $\mathbb{Q}$ with the holes filled.

The gap. That narrative quietly presumes a notion of “hole,” and a hole is only visible relative to a way of measuring closeness. Rudin uses the ordinary absolute value throughout and never says it is a choice.

Where the gap bites. Nowhere in this book — which is exactly why it is worth one page. A reader who never learns that the choice existed will later meet $\mathbb{Q}_p$ and find it inexplicable, and will also misread the word “complete” in Chapter 3 as a property of a *set* rather than of a *metric*.

What this section supplies. A second absolute value on $\mathbb{Q}$, and the different field it completes to.

S1.12 Definition. (*The p -adic absolute value.*) Fix a prime p . Every nonzero rational x is uniquely $x = p^v m/n$ with m, n coprime to p ; put $v(x) = v$ and

$$|x|_p = p^{-v(x)}, \quad d_p(x, y) = |x - y|_p.$$

Then d_p is a metric on $\mathbb{Q}$, and $(\mathbb{Q}, d_p)$ is not complete; its completion is the field $\mathbb{Q}_p$ of p -adic numbers.

SOURCE: Yu Pin p. 125, as a worked corollary of his completion theorem (Thm 77, pp. 123–124).

What “complete” is a property of

Under d_p a number is *small* when it is divisible by a high power of p . With $p = 5$, the numbers 5, 25, 125, ... march to zero, and $1/5$ is enormous. The metric is genuinely a metric — and it satisfies something stronger than the triangle inequality, $|x+y|_p \leq \max(|x|_p, |y|_p)$, which makes its geometry unrecognisable: every triangle is isosceles, and every point of a ball is its centre.

Complete $(\mathbb{Q}, d_p)$ and you get $\mathbb{Q}_p$, not $\mathbb{R}$. So:

“ $\mathbb{R}$ is $\mathbb{Q}$ with the holes filled” is *incomplete as a description*. The correct statement is that $\mathbb{R}$ is $\mathbb{Q}$ completed *with respect to the ordinary absolute value*. Change the absolute value and the holes move.

This is why Chapter 3 defines completeness for a metric space and not for a set. Rudin’s 3.11 is stated for metric spaces precisely because the property depends on d ; his Exercise 3.24 constructs the completion in general. The Chapter 2 appendix (S2.8) gives

the definition, and Yu Pin's universal property is what makes “the completion” legitimate language.

There is a theorem lurking here that neither book states: Ostrowski's, that the ordinary absolute value and the p -adic ones are the *only* absolute values on $\mathbb{Q}$ up to equivalence. So $\mathbb{R}$ and the $\mathbb{Q}_p$ are the complete list of completions — Rudin's choice was not arbitrary, merely unremarked.

NOT IN RUDIN

SUPPLEMENT G — REAL EXPONENTS, WORKED

WHY THIS SUPPLEMENT BELONGS TO CHAPTER 1

What Rudin does. Theorem 1.21 proves that every positive real has a unique positive n th root. Exercises 1.6 and 1.7 then ask the reader to build b^x for all real x and establish its laws, with a hint.

The gap. The exercises are substantially harder than they look, and two steps in particular are glossed by the hint: that $b^{m/n}$ does not depend on how the fraction m/n is written, and that the limit defining b^x for irrational x exists at all.

Where the gap bites. Every later appearance of x^α , a^x and $\log$ — 3.20's special limits, 8.6's exponential — rests on this exercise having been done. A reader who skipped it has a gap under the whole of Chapters 3 and 8.

What this section supplies. Tao's worked version of Rudin's exercise, with both glossed steps proved.

S1.13 Theorem. (*Rational exponents.*) For $x > 0$ and n a positive integer put $x^{1/n} = \sup\{y \geq 0 : y^n \leq x\}$. The set is nonempty and bounded above, so the supremum exists; it satisfies $(x^{1/n})^n = x$, and for integers a and positive integers b the definition $x^{a/b} = (x^{1/b})^a$ is independent of the representation of the fraction. With these, $x^{q+r} = x^q x^r$ and $(x^q)^r = x^{qr}$ for all rational q, r .

SOURCE: Tao I, Def. 5.6.4–5.6.7 and Lemmas 5.6.5–5.6.9 (bk. pp. 123–125) — Rudin's Exercise 1.6 worked out.

Proof. The set $\{y \geq 0 : y^n \leq x\}$ contains 0 and is bounded above by $\max\{1, x\}$, since $y > \max\{1, x\}$ forces $y^n \geq y > x$. So the supremum exists. Because $t \mapsto t^n$ is strictly increasing on $[0, \infty)$, the set is the interval $[0, y]$ where y is the unique positive n th root supplied by 1.21; hence $x^{1/n} = y$ and $(x^{1/n})^n = x$.

For independence, suppose $a/b = c/d$, so $ad = cb$. Both $(x^{1/b})^a$ and $(x^{1/d})^c$ are positive, and raising each to the power bd gives x^{ad} and x^{cb}

respectively — equal numbers. Since 1.21 provides *exactly one* positive bd th root, the two agree.

The laws $x^{q+r} = x^q x^r$ and $(x^q)^r = x^{qr}$ now follow by writing q and r over a common denominator and using the corresponding integer identities together with uniqueness in 1.21. $\square$

Rudin's 1.21 already gives the root; the content here is the *independence*, which needs the uniqueness half of 1.21 to force $(x^{1/bd})^{ad} = (x^{1/b})^a$ whenever $a/b = c/d$.⁶³

S1.14 Theorem. (*Irrational exponents.*) Let $x > 0$ and let α be real. If (q_n) is any sequence of rationals with $q_n \rightarrow \alpha$, then (x^{q_n}) converges, and the limit does not depend on the sequence chosen. One may therefore define $x^\alpha = \lim_n x^{q_n}$.

SOURCE: Tao I, Lemma 6.7.1 (bk. pp. 152–153) — Rudin's Exercise 1.7.

Proof. Take $x \geq 1$ (for $0 < x < 1$ apply the result to $1/x$). If q, r are rationals with $|q|, |r| \leq M$ and $|q - r| \leq 1/K$, then

$$|x^q - x^r| = x^r |x^{q-r} - 1| \leq x^M (x^{1/K} - 1),$$

and $x^{1/K} \rightarrow 1$ as $K \rightarrow \infty$ by 3.20(b). Now let $q_n \rightarrow \alpha$ with each q_n rational. The sequence (q_n) is Cauchy and bounded, so the displayed estimate makes (x^{q_n}) Cauchy, and it converges by completeness of $\mathbb{R}$ (3.11).

If (q'_n) is a second rational sequence tending to α , interleave the two into a single sequence; it also tends to α , so its image converges, and the two original limits — being limits of subsequences of a convergent sequence — coincide. $\square$

THE ESTIMATE THAT MAKES THE DEFINITION LEGAL

Everything rests on one inequality. For rationals q, r with $|q - r| \leq 1/K$ and both at most M in absolute value,

$$|x^q - x^r| \leq x^M (x^{1/K} - 1) \quad (x \geq 1),$$

and $x^{1/K} \rightarrow 1$ as $K \rightarrow \infty$ — which is Rudin's own 3.20(b) in disguise. So (x^{q_n}) is Cauchy whenever (q_n) is, and completeness of $\mathbb{R}$ supplies the limit. Interleaving two sequences with the same limit shows the value is independent of the choice, exactly as in the uniqueness argument of S1.1.

Two things are worth noticing. First, the definition needs completeness of $\mathbb{R}$, so real exponentiation is downstream of 1.19 — it could not have been introduced in $\mathbb{Q}$. Second, this is the same three-step pattern used throughout: *approximate by rationals, show the approximations are Cauchy, pass to the limit*. S1.1 uses it to build an isomorphism, S1.7 to build a supremum, and Chapter 4's S4.17 to extend a uniformly continuous function. Recognising the pattern is worth more than any one instance.

63 Rudin's hint for Exercise 1.6 says to prove $(b^m)^{1/n} = (b^p)^{1/q}$ when $m/n = p/q$ and then define $b^{m/n}$ to be that common value. That single sentence is the independence step, and it is where all the work is — both sides are n th roots, so uniqueness in 1.21 is what identifies them.

NOT IN RUDIN

APPENDIX EXERCISES

HOW THESE DIFFER FROM RUDIN'S TWENTY

Rudin's own Exercises 1.1–1.20 end the chapter proper. They are almost all verifications inside a structure already given: check a field axiom, compute with cuts, establish an inequality. These fifteen come from the comparison texts and ask two other kinds of question — what the axioms *cost*, and how to actually compute with them.

Group I (1–5): the toolkit. Drills on the four facts of Supplement D, plus the two supremum lemmas every later chapter uses and Rudin never assigns. Exercise 4 is the first place in this book where anyone is asked to *compute* a supremum.

Group II (6–10): what the axioms buy. Where each hypothesis is spent — induction failing on $\mathbb{Z}$, the supremum principle producing Archimedes, and the disconcerting question of whether one can prove any real numbers exist at all.

Group III (11–15): decimals and construction. Exactly which reals have two decimal names, the supremum built by hand, and a game.

Marked ★: harder, or requiring a result from a later chapter.

64 The ε -principle of S1.9 as a drill, and the single most-used move in the book. The proof is one line by contradiction — take $\varepsilon = |a - b|$ — but the habit it builds is what matters: to prove two things equal, show their distance is smaller than every positive number. Rudin uses this at 1.11, 1.21, 3.2(b) and throughout Chapter 3 without ever stating it.

65 Read each expression as a total distance to fixed points on the line. In (a) the sum of distances to 1 and 2 is least — and equal to the gap — exactly when x lies between them. In (b) the middle point is free to be hit exactly, so equality occurs only at $x = 2$: the minimiser of a sum of distances is the median. A geometric habit worth acquiring before Chapter 2 makes distance abstract.

66 Two right triangles sharing the leg a : the hypotenuses cannot differ by more than the other legs do. Algebraically it says $t \mapsto \sqrt{a^2 + t^2}$ never stretches distances — it is what Chapter 4 will call a Lipschitz function with constant 1. Rudin's Chapter 1 exercises contain nothing geometric; this one rewards a picture.

67 Rudin's Chapter 1 never asks anyone to compute a supremum, and computing four of them is the fastest way to internalise that Definition 1.8 has two clauses — an upper bound, and no smaller one. Note (b), where the set is unbounded above and the answer is $+\infty$ only in the extended sense of 1.23; and (c), where the supremum is 1 and is not attained, while every neighbourhood of it is crowded.

S1.1. Let $a, b \in \mathbb{R}$. Prove that if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$.⁶⁴

SOURCE: ECNU 习题 1.1 #3 (bk. p. 4).

S1.2. Prove that for all $x \in \mathbb{R}$: (a) $|x - 1| + |x - 2| \geq 1$; (b) $|x - 1| + |x - 2| + |x - 3| \geq 2$. Determine in each case exactly when equality holds.⁶⁵

SOURCE: ECNU 习题 1.1 #5 (bk. p. 4).

S1.3. Let $a, b, c > 0$. Prove $|\sqrt{a^2 + b^2} - \sqrt{a^2 + c^2}| \leq |b - c|$, and give the inequality a geometric meaning.⁶⁶

SOURCE: ECNU 习题 1.1 #6 (bk. p. 4).

S1.4. For each set, find sup and inf and verify the answer from Definition 1.8: (a) $\{x : x^2 < 2\}$; (b) $\{n! : n \in \mathbb{N}\}$; (c) $\{x \text{ irrational} : 0 < x < 1\}$; (d) $\{1 - 2^{-n} : n \in \mathbb{N}\}$.⁶⁷

SOURCE: ECNU 习题 1.2 #4 (bk. p. 8).

S1.5. Let A, B be nonempty and bounded, and $A + B = \{x + y : x \in$

$A, y \in B\}$. Prove $\sup(A+B) = \sup A + \sup B$ and $\inf(A+B) = \inf A + \inf B$.⁶⁸

SOURCE: ECNU 习题 1.2 #7 (bk. p. 8).

S1.6. Show that the principle of induction does not apply to $\mathbb{Z}$: exhibit a property $P(n)$ of integers such that $P(0)$ holds and $P(n)$ implies $P(n+1)$ for every integer n , yet $P(n)$ fails for some integer.⁶⁹

SOURCE: Tao I, Ex. 4.1.8 (bk. p. 81).

S1.7. Assuming only the field axioms and the order axioms, prove that the least-upper-bound property implies the archimedean property.⁷⁰

SOURCE: Yu Pin, homework A7 (p. 36).

S1.8. ★ Suppose $a > 0$. Can you prove that the open interval $(0, a)$ is nonempty?⁷¹

SOURCE: Yu Pin, 思考题 (p. 22).

S1.9. Prove that the ordered field $\mathbb{R}(t)$ of Supplement B really is ordered — that the sum and product of eventually-positive rational functions are eventually positive, and that trichotomy holds — and then verify directly that the set $\mathbb{N}$ has no least upper bound in it.⁷²

SOURCE: Supplied here; see the provenance note at S1.2.

S1.10. ★ Prove that, given the field and order axioms together with the archimedean property, the least-upper-bound property and the nested-interval property are equivalent. Then explain why the archimedean hypothesis cannot be dropped from this statement.⁷³

SOURCE: Yu Pin, Remark 4 (p. 28) and Thm 5 (pp. 26–27).

S1.11. Prove that a real number has two decimal representations if it is a terminating decimal — that is, if $x = n/10^m$ for integers n, m — and exactly one otherwise. Show also that the only decimal representations of 1 are $1.000\dots$ and $0.999\dots$ ⁷⁴

SOURCE: Tao I, Ex. B.2.2–B.2.3 (bk. p. 338); Yu Pin, homework E1 (p. 506), which states it as a four-way equivalence.

S1.12. Prove the supremum principle by the digit construction of S1.7 for a set of your own choosing — for instance $S = \{x : x^2 < 2\}$ — and carry the algorithm far enough to produce the first four digits of $\sup S$.⁷⁵

SOURCE: ECNU 定理 1.1 (bk. pp. 5–7).

68 The workhorse lemma behind every limit-algebra proof, and it is not in Rudin's exercise set. Both halves are needed: " $\leq$ " because $\sup A + \sup B$ bounds $A + B$, and " $\geq$ " because any smaller bound would fail for some x near $\sup A$ and y near $\sup B$ — which is the second clause of 1.8 doing its work. Compare 3.3, where the same statement for limits gets a whole theorem.

69 Take $P(n)$ to be " $n \geq 0$." One line, and a permanent lesson: induction is not a general logical principle but a specific axiom about $\mathbb{N}$, and what makes it work is that $\mathbb{N}$ has a starting point with nothing before it. Rudin uses induction freely throughout Chapter 1 — at 1.1, at 1.21, in the appendix — and never says what licenses it.

70 This is Rudin's Theorem 1.20(a), but posed inside an axiom system where Archimedes is independent, so that the exercise shows what is being bought rather than merely how. Compare S1.3, and note the shape of the argument: if $\mathbb{N}$ were bounded, $\sup \mathbb{N} - 1$ would fail to be an upper bound, and the integer beating it has a successor beating the supremum.

71 Disconcerting, and meant to be. On the field and order axioms alone the only elements you have names for are 0, 1 and -1 , so producing a real strictly between 0 and a is not the triviality it looks like — you must manufacture new elements, and manufacturing elements in bulk is precisely what the completeness axiom is for. Rudin's 1.19 hands the reader a fully populated $\mathbb{R}$ by fiat, which is why the question cannot even be asked inside his development. Yu Pin's own comment: "you will see the shadow of the Dedekind cut here."

72 The verification behind S1.2 and S1.3. For trichotomy, a nonzero rational function has finitely many zeros, so it has constant sign near $+\infty$. For the second part, show that if f bounded $\mathbb{N}$ above then so would $f - 1$, since $f - 1 > n$ whenever $f > n + 1$ — so no candidate can be least. The example is the one thing Chapter 1 lacks: a field where Archimedes visibly fails.

73 The precise form of Chapter 1's comparison box. One direction is a bisection; the other is the argument of S2.3 in the Chapter 2 appendix. For the last part, look at S1.2: nested intervals alone cannot rule out infinite elements, so it cannot imply Archimedes, whereas LUB implies both at once. This is exactly why Yu Pin lists four axioms where Rudin needs one.

74 The statement Rudin's 1.22 declines to make. He asserts that the expansion exists and is unique, having silently fixed the non-terminating digits by always choosing the largest digit at each stage; ECNU fixes the opposite convention (S1.4). The exercise is to see that the ambiguity is real, is confined to exactly the terminating decimals, and is always a choice between a 0-tail and a 9-tail.

75 The point is to watch Definition 1.8's two clauses act as loop invariants: at each stage one keeps the largest digit that some element still reaches. Doing it once on $\{x : x^2 < 2\}$ produces 1.414 and makes concrete what Rudin's 1.1 says is missing from $\mathbb{Q}$ — the digits are perfectly well defined at every finite stage, and only the infinite string fails to be rational.

76 The same construction used in S2.7 to recover sup from Cauchy completeness, here isolated. Uniqueness of m is where the archimedean property enters — it is what guarantees the set of integers k with k/n an upper bound is nonempty and bounded below. Compare with exercise 12: one construction walks in steps of 10^{-k} , the other in steps of $1/n$.

S1.13. ★ Construct the supremum of a nonempty bounded set by successive rational approximations, *without* invoking the least-upper-bound property: for each n find the unique integer m such that m/n is an upper bound while $(m - 1)/n$ is not, and prove m is unique.⁷⁶

SOURCE: Tao I, Ex. 5.5.2–5.5.3 (bk. p. 121).

S1.14. Prove that $b^{m/n}$ is well defined — that $(b^p)^{1/q} = (b^m)^{1/n}$ whenever $m/n = p/q$ — and deduce the law $b^{r+s} = b^r b^s$ for rational r, s .⁷⁷

SOURCE: Tao I, Lemma 5.6.8 (bk. p. 124); Rudin's Ex. 1.6.

S1.15. ★ (*A game.*) Two players alternately choose nested closed intervals $I_1 \supset I_2 \supset \cdots$, each at most half the length of its predecessor, so that the intersection is a single point x . The first player wins if x is rational, the second if x is irrational. Who has a winning strategy?⁷⁸

SOURCE: Yu Pin, problem G (p. 69), the Banach–Mazur game.

77 Rudin's Exercise 1.6, and the independence step is the whole of it. Both sides are positive n qth roots of b^{mq} , so the uniqueness half of 1.21 identifies them — one more instance of the pattern flagged at 1.21, where an existence-and- uniqueness theorem is used entirely for its uniqueness. See S1.13.

78 Everything needed lives in Chapter 1: nested intervals, density of $\mathbb{Q}$, and the uncountability that follows from S2.17 in the Chapter 2 appendix. The second player wins, and the reason is the same reason the diagonal argument works — the first player must commit to a countable list of targets while the second retains room to dodge each one in turn. A pleasant way to see that “most” reals are irrational without ever using the word measure.